

NBER WORKING PAPER SERIES

THE COV-LITE LIQUIDITY ADVANTAGE, REGULATORY PRESSURES, AND
THE EVOLUTION OF THE LEVERAGED LOAN MARKET

Robert Prilmeier
René M. Stulz

Working Paper 35617
<http://www.nber.org/papers/w35617>

NATIONAL BUREAU OF ECONOMIC RESEARCH
1050 Massachusetts Avenue
Cambridge, MA 02138
August 2026

We are grateful for excellent research assistance from Andrei Gonçalves, Byungwook Kim, Leandro Sanz, and Kris Shen. The views expressed herein are those of the authors and do not necessarily reflect the views of the National Bureau of Economic Research.

NBER working papers are circulated for discussion and comment purposes. They have not been peer-reviewed or been subject to the review by the NBER Board of Directors that accompanies official NBER publications.

© 2026 by Robert Prilmeier and René M. Stulz. All rights reserved. Short sections of text, not to exceed two paragraphs, may be quoted without explicit permission provided that full credit, including © notice, is given to the source.

The Cov-lite Liquidity Advantage, Regulatory Pressures, and the Evolution of the Leveraged Loan Market

Robert Prilmeier and René M. Stulz

NBER Working Paper No. 35617

August 2026

JEL No. D82, G18, G23, G32

ABSTRACT

Following the global financial crisis (GFC), regulators made it harder for banks to retain leveraged loan exposure. We conjecture their actions increased the share of leveraged loan issuances with no maintenance covenants (cov-lite loans) because such loans are easier to sell. We find that, post-GFC, the share of cov-lite loan issuance increased more for banks facing stricter regulation, failing to pass a stress test outright, or exhibiting greater vulnerability to the severely adverse stress test scenario. We show that, as expected from theory, cov-lite loans have a liquidity advantage that lowers their credit spread and is higher for private firms.

Robert Prilmeier
Tulane University
rprilmei@tulane.edu

René M. Stulz
The Ohio State University
and NBER
stulz@cob.osu.edu

1. Introduction

Since the global financial crisis (GFC), the market for loans to risky corporations (leveraged loans) has undergone a dramatic transformation. The typical bank-intermediated leveraged loan has a revolver tranche held by banks and a term B tranche held by institutional investors. In 2007, less than 20% of leveraged loans included a term B tranche with no maintenance covenants. Maintenance covenants are financial covenants that require the borrower to maintain certain financial ratios through the life of the loan. Loans without maintenance covenants are typically called cov-lite loans. In 2021, 84% of leveraged loan facilities underwritten by banks included a cov-lite loan. Paradoxically, this evolution is even more dramatic for loans by private companies, which would seem riskier, than it is for loans by public companies. In 2021, 94% of the leveraged loans to private companies were cov-lite loans. At the same time, the private credit market, a market with no bank intermediation where loans typically have maintenance covenants, has grown dramatically (Block, Jang, Kaplan, and Schulze, 2024; Chernenko, Ialenti, and Scharfstein, 2025; Jang, 2025).

In this study, we provide and test a risk management explanation for the rise of cov-lite loans. After the GFC, regulators increased capital requirements on banks holding riskier loans, introduced stress tests, and gave guidance that discouraged banks from holding riskier loans. The guidance emphasized the importance of pipeline risk, the risk that syndicated loans expected to be sold to investors would be stuck on bank balance sheets, so that it gave incentives to banks to issue loans that would be likely to have a liquid market. This evolution made it more costly for banks to have illiquid risky loans on their balance sheet, thus incentivizing banks to issue loans that are likely to be more liquid.

Theory predicts that loans with maintenance covenants are less liquid than loans without such covenants (see Parlour and Plantin, 2008, for a model of loan liquidity). The reason for this lies in an adverse selection problem. Investors who hold loans with maintenance covenants acquire information in the process of monitoring covenants. As a result, an investor considering the purchase of a loan from its current holder is at an informational disadvantage. To compensate for that informational disadvantage, the investors will

want better terms because the loan with maintenance covenants will be more expensive to trade. For the same risk, loans with maintenance covenants will have a higher interest rate than loans without such covenants, thus partially negating the cost advantage maintenance covenants offer borrowers in the traditional lending model. Combining the regulatory incentives with this theory of loan liquidity, we expect banks to shift toward issuing a greater share of cov-lite loans relative to cov-heavy loans.

High-yield term loans are mostly held by institutional investors directly or through collateralized loan obligations (CLOs) (Lee, Li, Meisenzahl, and Sicilian, 2019). Investors who hold high-yield loans have many reasons to want these loans to be relatively liquid. CLOs are the largest investors in high-yield loans followed by mutual funds. Typically, CLO managers value loan liquidity (Fleckenstein, 2022). CLO managers can trade loans in a CLO for several years early in the life of the CLO. They hope to create value through trading. Any gains realized through trading loans are especially valuable to the holders of the equity tranche of the CLO, which is generally held by the manager of the CLO. Mutual fund managers hope to create value through trading as well. Further, however, mutual funds at times have to cope with large redemptions, which force them to sell assets.

While investors value the liquidity of loans, banks have good reasons to value loan liquidity as well and do so more after the GFC than before. There are at least four reasons why banks are more motivated to prefer liquid loans after the GFC. First, many loans were stuck on bank balance sheets because of the GFC and banks suffered losses on these loans (Ivashina and Scharfstein, 2010; Benmelech and Bergman, 2011). Second, risk-based capital requirements increased substantially for leveraged loans held on bank balance sheets after the global financial crisis. More generally, balance sheet usage became more expensive after the global financial crisis (Duffie, 2022). As a result, banks' incentives to underwrite loans that can be moved more easily off their balance sheets increased. Third, bank regulators and policymakers expressed concerns about high-yield loans. The Federal Reserve issued guidance on highly leveraged loans for banks in 2013 and leveraged lending was mentioned in the minutes of 14 of 16 FOMC meetings in 2013 and 2014 (Nelson, Baer, and Newell, 2017). As a result, bank examiners appear to have paid more attention to and been more critical of these loans. To avoid criticisms, banks again have incentives to make loans they can

easily move off their balance sheets. Lastly, banks generally are committed to making a market for loans they have sold off to institutional investors. This means that loans may come back to the balance sheet of the bank. The bank is more likely to be able to sell loans that come back to its balance sheet if these loans are liquid. To the extent that regulators and supervisors are concerned about such loans, banks have incentives to make sure that these loans do not get stuck on their balance sheet.

We first investigate whether the regulatory changes following the GFC led banks to issue relatively more cov-lite loans. Kim, Plosser, and Santos (2018) study the lessons from the leveraged lending guidance issued by the Federal Reserve, the FDIC, and OCC in March 2013. They focus on the level of lending rather than the composition of lending, which is our focus. The guidance was clarified in late 2014. They find that the guidance, after its clarification, adversely affected the volume of lending for the institutions overseen by the Large Institution Supervision Coordinating Committee (LISCC). The LISCC institutions are the most closely supervised institutions, so that they would be expected to follow the guidance most closely. We use their difference-in-differences approach and estimate the treatment effect of the guidance on the composition of leveraged lending for the LISCC institutions. We find that relative to non-LISCC institutions, the LISCC institutions increase the proportion of leveraged loans without maintenance covenants. Furthermore, we find that the increase in the cov-lite proportion of leveraged loan issuances is concentrated in riskier loans for LISCC institutions, but not for non-LISCC institutions.

We complement the leveraged lending guidance analysis with a second, separate regulatory shock: banks' stress-test outcomes. We expect banks viewed by regulators as having too much risk to reduce their issuance of leveraged loans with maintenance covenants relative to other banks. This additional exercise strengthens identification for two reasons. First, it demonstrates that our empirical predictions hold across two distinct shocks rather than hinging on a single policy episode. Second, while we do test the parallel trends assumption, the leveraged lending guidance affects all banks at the same time, which leaves open the possibility that time trends diverge across large and smaller banks for reasons other than regulation. By contrast, stress-test outcomes are bank-specific and occur at staggered dates across otherwise comparable large commercial banks. This relative homogeneity and staggered timing allow us to absorb the market-

wide trend in cov-lite issuance with time fixed effects and isolate the bank-level response. Using this design, we find that banks that fail to pass a stress test without reservations subsequently increase the share of leveraged loans issued without maintenance covenants. Finally, we use quantitative results from the Dodd-Frank stress tests (DFAST) to quantify banks' need for capitalization resilience. We find that banks whose capitalization is more vulnerable to a severely adverse shock originate a larger share of leveraged loans in the form of cov-lite loans.

Maintenance covenants have the benefit that they make a loan safer for the lender (Bradley and Roberts, 2015; Matvos, 2013). This is because the lender can monitor the loan through the covenants and force the borrower to repay the loan or take actions that make the loan safer.¹ However, as already discussed, the monitoring of covenants creates an adverse selection problem. The information environment of the borrower affects the magnitude of the adverse selection problem. When the borrower is a public company, investors have access to much more information about the company. This richer information environment reduces the adverse selection problem. The adverse selection problem created by covenants is worse for private firms than for public firms because of the weaker information environment of private firms. As a result, the liquidity benefit of cov-lite loans should be greater for private firms than for public firms. We would therefore expect, for a given level of loan risk, cov-lite loans to be more liquid than cov-heavy loans (the liquidity hypothesis). For a given level of loan risk, we would expect the liquidity advantage of cov-lite loans to be higher for loans issued by private firms than by public firms (private firm liquidity advantage of cov-lite loans hypothesis). Finally, for a given level of loan risk, we would expect cov-heavy loans to have a higher yield than cov-lite loans because investors in cov-lite loans do not have to monitor the loan through covenants and receive the benefit of higher liquidity (the cov-lite yield advantage hypothesis).

We first test the liquidity hypothesis and the private firm liquidity advantage of cov-lite loans hypothesis. All our results are obtained using industry, year-quarter, and rating fixed effects. We find that cov-lite loans are more likely to begin trading than cov-heavy loans. The time-to-break, the time from the

¹ That said, Bräuning, Ivashina, and Ozdagli, 2025 show that cov-lite loans still have incurrence covenants and incurrence covenants have bite when their threshold is crossed.

launch date to the time the loan starts trading, is shorter for cov-lite loans. Once leveraged loans start trading, cov-lite loans trade more. Finally, these loans have lower bid-ask spreads and more quotes. A key assumption in our tests is that we can measure a loan's risk through ratings. We verify the validity of this assumption by examining the performance of loans and find no difference in bankruptcy risk for cov-lite vs cov-heavy loans.

When we turn to the hypothesis that the liquidity advantage is higher for cov-lite loans issued by private firms, we find support for the hypothesis for all our metrics except one. Specifically, we do not find that cov-lite loans issued by private firms are more likely to trade than cov-lite loans issued by public firms. However, we show that, relative to cov-heavy loans issued by private firms, cov-lite loans issued by private firms enter the secondary market faster, have a higher percentage of the loan traded, have lower bid-ask spreads, and have more quotes. It follows from these results that we find support for the hypothesis that cov-lite loans are more liquid than cov-heavy loans and that the liquidity advantage of cov-lite loans over cov-heavy loans is greater for loans to private firms than to public firms for a given industry, year, and rating.

Next, we test the credit spread hypothesis using the credit spread of the institutional term loan tranche as the dependent variable. The existing literature diverges on whether cov-lite loans have higher credit spreads (e.g., Billett, Elkamhi, Popove, and Pungaliya, 2016) or lower on average (Badoer, Emin, and James, 2024, and Becker and Ivashina 2025). Our results are again obtained using industry, year-quarter, and rating fixed effects, and controlling for an array of deal and firm characteristics. To further control for differences in credit risk across loans, we add as a control variable the spread of the revolver tranche within the same deal. Since the revolver prices credit risk, the difference in spreads across cov-lite and cov-heavy loans after controlling for the revolver spread should capture investor preferences for holding cov-lite loans or cov-heavy loans. We find that, after controlling for risk, cov-lite loans have lower spreads than cov-heavy loans. We depart from the literature by testing hypotheses related to the difference between private and public borrowers and to the role of liquidity. Turning to the hypothesis that private firms enjoy a greater yield advantage from cov-lite loans, we find that cov-lite loans issued by private firms have lower spreads

relative to cov-heavy loans than do cov-lite loans issued by public firms. Finally, we connect the liquidity advantage to the yield advantage by showing that the loan's bid-ask spread is priced in the loan's yield. This result is robust to instrumenting the loan's bid-ask spread using the industry-level bid-ask spread at the time of origination while purging industry fixed effects and rating x time fixed effects.

Becker and Ivashina (2025) document the importance of coordination costs in renegotiation for the popularity of cov-lite loans. The argument is that renegotiation requires coordination and coordination is difficult and time-consuming. Cov-lite loans have lower coordination costs than cov-heavy loans because they do not require coordination around renegotiation of covenants. The lower coordination costs of cov-lite loans are a source of their greater liquidity. Coordination costs directly affect the liquidity of a claim in the following way. Some investors have a comparative advantage in dealing with coordination issues. They may have an infrastructure in place or knowledgeable employees. Other investors may find it extremely costly to deal with coordination costs. Loans subject to low coordination costs can be held and traded by both types of investors. In contrast, loans with high coordination costs typically are held more or only by investors that have a comparative advantage in dealing with coordination issue. As a result, the loans with more coordination issues are expected to be less liquid because fewer investors are willing to hold such loans. The adverse selection problem and the greater coordination costs of cov-heavy loans both predict greater liquidity of cov-lite loans. We do not attempt to distinguish between these two reasons for the greater liquidity of cov-lite loans.

An important implication of our study is that regulators' concerns about the leveraged loan market had the unintended effect of reducing banks' willingness to lend to highly leveraged firms with maintenance covenants. The literature suggests that firms use substitute mechanisms to reduce the risk of loans when a loan does not have maintenance covenants, either by including a bank-held covenant-heavy tranche in the loan package (Berlin, Nini, and Yu, 2020) or by relying on the reputation of a sponsor (Badoer, Emin, and James, 2023). Almost all highly-leveraged loans to private firms have a sponsor. While private equity sponsors' reputation may act as a partial substitute for covenants, it seems plausible that riskier firms can still benefit from such covenants in that they can reduce the risk of their loans. Some firms might not be

able to borrow at all without such covenants. The role of banks in issuing risky loans with maintenance covenants has been replaced in part by private credit. Private credit loans are illiquid and involve extensive use of covenants (Jang, 2025).

Our paper contributes to several literatures. First, we add to the large literature on the role of covenants and on why loans may not have maintenance covenants. The literature on covenants has mostly focused on loans that stay on banks' balance sheets. However, most highly leveraged loans do not stay on bank balance sheets. Our paper shows that analyses of the role of covenants must account for their impact on loan liquidity. In the syndicated loan market, investor demand for liquidity in the secondary market influences the characteristics of loans in the primary market. Second, our paper contributes to the literature on highly leveraged loans. There is much concern among regulators and policy makers about the risks associated with these loans. We show that regulation has impacted the composition of this market and that market structure is in part determined by banks' risk management requirements. Third, we add to the banking literature on the originate-to-distribute model of lending. We show that loan attributes that reduce risk when banks hold these loans may make them more difficult to sell. Fourth, we add to the literature on funding for private firms. We review the existing literature on cov-lite loans and our contribution to it in the next section.

After presenting our data in section 3, we document the evolution in the market for leveraged loans that motivates our paper in Section 4. In Section 5, we show how regulatory changes led to the growth of the share of cov-lite loans in the leveraged loan market. In Section 6, we show that cov-lite loans are more liquid than cov-heavy loans and that the liquidity advantage of cov-lite loans is greater for private firms, as one would expect from theory. In Section 7, we examine how interest rates differ between cov-lite loans and cov-heavy loans. We conclude in Section 8.

2. The cov-lite literature

In this section, we briefly review the cov-lite literature. We focus on the papers that motivate subsequent analyses.

Ayotte and Bolton (2011) model the choice between cov-heavy and cov-lite loans. In their model, loans are securitized, so that investors have to expend reading costs to learn about loans. With a cov-lite loan, reading costs are low, but investors assume that any terms they do not know are adverse to them. With a cov-heavy loan, reading costs are high, but investors have lower moral hazard costs. In good times, expected moral hazard costs are lower, so that cov-lite loans are more advantageous. In their model, cov-lite loans are more liquid than cov-heavy loans because of lower reading costs. The model predicts pro-cyclical issuance of cov-lite loans. It also predicts that cov-lite loans have higher credit spreads and are more likely to be securitized.

Billett, Elkamhi, Popov, and Pungaliya (2016) use a different approach based on the existence of two agency problems with loans. One problem is between the borrower and the lender; the other problem is between the bank and the non-bank investors. In their model, covenant tightness increases as the bank participation in a loan falls, but only up to some threshold level of bank participation. The logic for this result is that the bank's incentive to monitor falls as it holds a smaller share of the loan. An increase in covenant tightness increases the bank's incentive to monitor. However, if the bank's skin in the game is too low, it may be optimal for the loan to be a cov-lite loan as the monitoring costs may be too high compared to the benefits of monitoring for some types of loans. The bank's skin in the game will be pushed down if there is a lot of investor demand for loans or if the cost of capital for banks is high. In their model, cov-lite loans have higher spreads than cov-heavy loans. The authors find that the value-weighted spread on a deal involving a cov-lite loan is higher and that loans sponsored by private equity groups are more likely to be cov-lite loans because these groups have reputation incentives that reduce moral hazard risks for investors.

Becker and Ivashina (2025) test three explanations for the growth of cov-lite loans. The first explanation is that borrowers, following the crisis, learned that renegotiation costs are high and are willing to pay to obtain unmonitored loans. The second explanation is that coordination costs for monitoring when a loan is held by many institutional investors are high, so that it can be optimal to have an unmonitored loan. The third explanation is that investors are chasing yields, so that they would rather have a higher yield with fewer protections. They find results that are strongly supportive of the coordination cost hypothesis.

Berlin, Nini, and Yu (2020) examine 946 loan contracts. They show that highly leveraged loan packages often have term A and revolver tranches that have maintenance covenants. In the case of revolvers, these maintenance covenants may be springing covenants, namely covenants that come in force once the borrower has drawn down the revolver. Where the loan package itself does not contain a monitored tranche, the firm often still has another loan with maintenance covenants in its capital structure. As a result, loans that have cov-lite tranches are often monitored by banks. Essentially, cov-lite packages put the lenders with a comparative advantage in monitoring, namely the banks, in charge of renegotiation.

Split-control loan deals may still have less monitoring than deals where banks have more skin in the game and in such deals banks may have incentives that are imperfectly aligned with those of the investors. The bank may be more concerned about protecting the value of the tranches with control rights that it owns than the tranches with no maintenance covenants. However, the literature points out that even when a bank's ownership of a loan package is low, its skin in the game may effectively be larger than it seems because it may have other interactions with the borrower. Neuham and Saidi (2016) point to non-loan exposures from universal banks and Jiang, Kundu, and Xu (2022) argue that banks have incentives to monitor because of the present value of fees they will earn from a loan.

Diamond, Hu, and Rajan (2019) develop a theory where in boom times firm leverage depends mostly on the value of firm assets as debt can be repaid out of the proceeds from selling firm assets. In contrast, when corporate assets become less liquid, loans are repaid out of cash flows and the pledgeability of cash flows becomes critical. Maintenance covenants increase pledgeability of cash flows, so that we would expect these covenants to be more important in periods of low liquidity. This theory suggests that covenant-lite loans are more likely to be issued in periods of high liquidity. The evidence in Becker and Ivashina (2025) supports the conclusion that the costs and benefits of issuing cov-lite loans vary depending on business conditions as the issuance of cov-lite loans essentially went to zero during the GFC.

With a cov-heavy loan, lenders monitor the borrower and hence have information that outsiders do not have. As a result, an adverse selection problem arises that can make the market for a loan less liquid or illiquid as shown in Parlour and Plantin (2008) in the context of loan sales. In contrast, the lack of

maintenance covenants with cov-lite loans reduces the adverse selection problem that results from the lenders being better informed and hence increases the liquidity of the loans. In addition, monitoring is costly and institutional investors may not have the technology to monitor effectively. The cov-lite structure of the institutional tranche amounts to a pre-commitment not to acquire private information and not to monitor. Such a pre-commitment makes the cov-lite tranche less informationally sensitive to private information than a cov-heavy tranche (Gorton and Pennacchi, 1990; Dang, Gorton, and Holmstrom, 2019).

Berlin, Nini, and Yu (2020) show that one solution to reduce the risk of cov-lite loans is to have a monitored tranche in the loan package. Cov-lite loans can have other mechanisms to reduce their risk. One mechanism is to have shorter maturity. Another mechanism is to have a sponsor whose reputation helps control the risk. Badoer, Emin, and James (2023) show that cov-lite loans have similar recovery rates and significantly lower spreads than loans with maintenance covenants. They provide evidence that the reputational capital of a loan's sponsor plays an important role in reducing the risk of cov-lite loans. Though not addressing cov-lite loans directly, Haque, Mayer, and Wang (2025) show that bank monitoring decreases when a loan has a private equity sponsor. This is because the private equity sponsor has reasons of its own to monitor and facilitate loan performance of the issuer. They show that the decrease in monitoring of the lead bank when there is a private equity sponsor has no performance cost. Badoer, Emin, and James (2023) show that the reputation of the private-equity sponsor is related to the performance of loans.

3. The data

Our leveraged loan sample includes 7,030 loans by US issuers on the US leveraged loan market from 2010 through 2021.² Our main data source is Capital IQ Leveraged Commentary and Data (LCD). The LCD database provides information on loan characteristics such as the size, yield spread, and maturity of each loan facility (for the advantages of LCD over DealScan for leveraged loans, see, e.g., Brauning,

² We exclude financial firms and utilities from our sample.

Ivashina, and Ozadagli, 2025). Importantly, it identifies covenant-lite facilities and second lien facilities. Unlike data available to academic researchers in DealScan, the LCD data contain information on the borrower's first lien bank loan rating along with the corporate credit rating. We limit our sample to originations and exclude amendments. Following Billett et al. (2016), we classify all leveraged loans that contain a covenant-lite facility as covenant-lite.³ The remaining leveraged loans are referred to as covenant-heavy.

Several of our analyses require matching LCD loan observations to DealScan. We match loans based on the identity of the borrower, identity of the lead arranger, approximate origination date,⁴ as well as the size, yield spread, and maturity of the loan tranches. We require substantial overlap in tranche characteristics, but do not require all characteristics to match since the two databases may record the same loan differently. For example, DealScan contains tranche type classifications that are not reported as separate types in LCD, such as bridge loans or delay draw term loans, and whose mapping can be ambiguous. We are able to match 93% of LCD borrowers and 72% of LCD loan observations to DealScan. The most frequent reason for a failure to match is that, for about 20% of LCD loan observations, DealScan does not record any loan observation for that borrower within 180 days after the LCD launch date.

When possible, we match borrowers to Compustat based on name and date. Note that for a borrower to be in Compustat, that borrower must have some security that was registered with the SEC at issuance, so that most private borrowers will not have a match in Compustat. To match borrowers in Compustat, we make use of the *spedis* function in SAS along with manual verification of similar names in Capital IQ. For loans that match to DealScan, we also obtain matches to Compustat from the link file provided by Chava and Roberts (2008). Compustat borrowers are matched to Capital IQ using GVKEY. We perform another name match to link non-Compustat borrowers to Capital IQ since this database includes privately held

³ While covenant-lite loan facilities are generally structured as term loans B, roughly half of the deals that contain a covenant-lite facility also contain a revolving line of credit.

⁴ Origination dates cannot be matched exactly since LCD reports the date on which a loan launches for syndication, while DealScan reports the date on which the deal closes, with launch dates reported only for a subset of the deals.

firms. Of the 3,160 distinct borrowers in the leveraged loan sample, we find 55% in Compustat and 99% in Capital IQ. Bankruptcy announcement dates are obtained from Capital IQ.

Two databases are used to gauge the liquidity of the leveraged loans in the secondary trading market. The CRSP mutual fund database provides information on loan shares held by loan participation mutual funds on a monthly basis along with mark-to-market prices as of the reporting date. Changes in shares held from month to month allow us to infer that a given mutual fund traded shares in the loan. We match loans to the CRSP mutual fund database using the security name indicated in CRSP. These data are consistently available from June 30, 2010 onwards.

We obtain secondary market loan pricing data from the Thomson Reuters/LSTA Mark-to-market pricing service. These data are available to us for the period of January 2010 through May 2023. We calculate bid-ask spreads as the difference between the average bid quote and the average ask quote divided by the average ask quote multiplied by 100. As a measure of market depth, the data also report the number of brokers providing a quote.

4. The changing composition of funding for highly leveraged firms

We now turn to the evolution of the market for leveraged loans and of the percentage of cov-lite loans. In contrast to the literature, we differentiate between private firm borrowers and public firm borrowers. The reasons for this are straightforward. First, our argument for why cov-lite loans are more liquid is an adverse selection argument. This argument is stronger for private firms than public firms because the information environment of private firms is sharply different from the investment environment of public firms. Second, the amount of cov-lite borrowing by private firms over our sample period is almost the same as the amount of cov-lite borrowing by public firms, so the two are of equal importance. The same is not the case for cov-heavy borrowing, as borrowing by public firms is more than twice the amount of borrowing by private firms.

We define a public firm as one that has at least one SEC-registered security outstanding. Table 1 shows data on aggregate leveraged loan issuance and issuance of cov-heavy and cov-lite loans. Except for a GFC collapse, the dollar amount of loans stays fairly stable over time except for an extremely high amount in 2021. The total amount of loans issued is \$407 billion in 2007. Total loan issuance volume exceeds that amount in four years since 2007. Before 2021, the highest amount is \$430 billion in 2018. The cov-heavy share of the issuance amount falls fairly steadily after the GFC after reaching a peak of 100% in 2008 to 16% in 2021. The cov-lite percentage of the issuance amount naturally increases since 2008, but it was 32% in 2007.

In Panel A of Figure 1, we show the percentage of loans originated by private borrowers, both in terms of loan number and loan amounts. Prior to the GFC, private borrowers accounted for about 40-50% of the number of loans issued. After a brief drop to about 35% during the crisis, the percentage returned to pre-crisis levels in 2010 and stayed there until it accelerated to 50-60% beginning in 2016. By contrast, the percentage of the aggregate loan amount originated by private firms has increased throughout the sample period, beginning at about 25% in 2005 and ending at about 55% in 2021.

In Panel B of Figure 1, we show the percentage of both the number and value of packages that are cov-lite for each year. We show these percentages separately for public firms and private firms. For all percentages, there is a sharp increase from 2011 to 2013. The percentage of cov-lite loans by volume for public firms is above 50% every year after 2012 except for 2020. However, this percentage exceeds 60% only in 2021, when it peaks at 69%. In contrast, the percentage of cov-lite loans for private firms exceeds the peak for public firms every year after 2015. The peak percentage of cov-lite loans for private firms is 94% in 2021.

Another way to see the sharp difference in the issuance of cov-lite loans between private and public firms is to look at the average percentage of cov-lite loans over the last five sample years. From 2017 to 2021, the average percentage of cov-lites is 71% for the sample as a whole. Turning to public firms, this average is 58%. In contrast, the average is 87% for private firms, or 50% higher for private firms compared to public firms.

5. Lending response to regulatory changes

We now turn to examining the effect of regulatory changes on the use of cov-lite loans in the market for leveraged debt. We use a difference-in-differences regression model for two different types of events. First, we examine shifts in market participants' cov-lite lending around the leveraged lending guidance of 2013 and differentiate between more and less strictly regulated banks, and, in a triple-differences setting, between riskier and less risky loans. Second, we study whether banks that fail to unconditionally pass a stress test increase their cov-lite lending to make it easier to move risky loans off their balance sheets. While neither of these approaches is perfect, we note that the first approach is driven by differences in the time series for different groups of banks, while the second approach allows for staggered differences across individual banks subject to the same regulatory regime, and thus any vulnerabilities of these approaches appear to be orthogonal to each other. Our third approach exploits the capital vulnerability of banks as measured in DFAST. We examine whether more capital vulnerable banks originate more cov-lites in the subsequent year.

5.1. Leveraged lending guidance

On March 30, 2012, the Office of the Comptroller of the Currency (OCC), the Federal Reserve Bank, and the Federal Deposit Insurance Corporation (FDIC) issued a proposed joint guidance on leveraged lending with a request for public comment. The proposal highlighted the growth of the leveraged loan market along with increased participation of unregulated investors and underwriting standards and capital structures deemed aggressive by the agencies. To counter these trends, the proposed guidance put banks on notice that they were expected to develop sound underwriting standards with consideration for appropriate leverage levels and the borrower's ability to repay, a consistently applied definition of leveraged finance, and strong risk management including stress testing, information systems, and pipeline management.

The prescriptions laid out in the guidance were vague (with the exception of certain limits on Total Debt/EBITDA and minimums on the repayment ability of borrowers) and subject to interpretation by banks

and examiners. Nonetheless, they put banks on notice that their levered loan portfolios and exposures would be under greater scrutiny going forward. They warned banks against poorly underwritten transactions and emphasized that underwriting standards should consider covenant protections among other things. Of note, the guidance highlighted the importance of secondary market liquidity for a bank's ability to sell down pipeline exposures. The guidance was finalized and published on March 21, 2013. In response to industry questions, on November 7, 2014, the agencies published a Frequently Asked Questions document clarifying the guidance. Among other things, the FAQ (Question 11) stated that cov-lite loans were not considered non-pass credits so long as the loans had other mitigating factors to ensure appropriate credit quality.⁵ While exposure to leveraged borrowers through trading assets had to be tracked under the guidance, trading desks were allowed to buy and sell non-pass credits, thus allowing banks to offload exposure so long as the loan could be traded in a sufficiently liquid secondary market. Finally, in March of 2018, the then-head of the OCC stated that banks no longer needed to adhere to the guidance. However, the OCC and the FDIC rescinded the guidance formally only in December 2025.⁶

Because the leveraged lending guidance highlighted examiners' concerns about banks' exposures to risky loans and their ability to sell down pipeline exposures, we hypothesize that the guidance increased incentives for banks to originate loans that are easier to sell and that these incentives were greater for the more closely supervised banks. Following Kim, Plosser, and Santos (2018), we therefore adopt a difference-in-differences design exploiting the fact that LISCC institutions are more heavily scrutinized by regulators than other banks. Specifically, we test the response of LISCC institutions relative to other institutions to

⁵ Despite this, Abuzov, Herpfer, and Steri (AHS, 2025) view the guidance as strongly opposed to cov-lite issuance by banks and argue that the guidance clarification caused a decrease in cov-lite issuance by banks and an increase in non-banks on the basis of a difference-in-differences approach where the guidance is treatment. Our evidence holds if we use the guidance clarification as treatment except that we find evidence of pre-trends associated with the guidance that differ across LISCC and non-LISCC institutions. Note that a non-bank in AHS is GE Credit. GE Credit is a LISCC bank at the time of the guidance and guidance clarification enactments. Our results hold if we remove GE Credit from the LISCC banks.

⁶ See "OCC and FDIC Rescind Interagency Leveraged Lending Guidance," Sullivan and Cromwell, December 9, 2025.

each of the treatment events outlined above: proposal, finalization, and clarification of the leveraged lending guidance as well as the OCC statement on non-enforcement.⁷

In Figure 2, we show the fraction of loan amounts that are cov-lite over time for LISCC and non-LISCC institutions. LISCC institutions generally originate a greater fraction of their loans as cov-lites than non-LISCC institutions. Consistent with regulatory scrutiny motivating banks to originate more loans as cov-lites, we note a rise in cov-lite issuance by all types of institutions beginning in 2012, around the time the leveraged lending guidance was proposed. Further consistent with regulatory incentives, the increase in cov-lite issuance by LISCC institutions appears to substantially outpace the growth in cov-lite lending among non-LISCC institutions.

In Table 2, we formally test for a difference between the responses of LISCC and non-LISCC institutions to leveraged lending guidance events. Specifically, using a panel of institution-quarter observations, we regress the cov-lite share of the total amount of leveraged loan issuance (cov-lite share) for each institution on event time indicators and their interactions with the institution's LISCC status. The sample covers 2010-2021. Standard errors are clustered by lead agent and launch quarter. To test for a possible pre-trend, we include an indicator variable that equals one during the four quarters prior to the proposal of the guidance. We do not find evidence of a pre-trend either for all institutions or for LISCC institutions. Column (1) shows that after the proposal of the guidance, the cov-lite share grows by 7 percentage points relative to 2010, and this growth increases to 14 percentage points after finalization of the guidance and 33 percentage points after clarification. The cov-lite share among all institutions continues to grow after the OCC statement that banks no longer had to follow the guidance. Column (1) also shows an immediate jump in the cov-lite share for LISCC institutions of 28 percentage points over and above the growth among non-LISCC institutions after the proposal of the guidance. This wedge grows to 38 percentage points after finalization of the guidance and remains significant after clarification of the guidance. It is not significant after the OCC statement. In Column (2), we test whether institutions that are

⁷ Our approach differs slightly from Kim, Plosser, and Santos (2018) in that they focus on two of the events studied here, namely the finalization and clarification of the guidance.

not LISCC institutions but are covered by the Comprehensive Capital Analysis and Review (CCAR) differ from banks that are covered by neither, but we do not find a difference for these banks. In Column (3), we drop loans made by investment banks and nonbanks and find very similar results. Finally, in Columns (4) through (6), we add lead agent fixed effects. Results are unchanged. Note that in Columns (3) through (6) the coefficient on the OCC statement for LISCC banks becomes positive and significant, so that the wedge for these banks does not change after the statement.

If regulatory concerns are driving the share of cov-lite originations, we would expect that the leveraged lending guidance would increase cov-lite originations in particular for riskier loans that are more likely to become non-pass credits, and we would expect this growth to be stronger for LISCC institutions. Almost all cov-lite loans have either a BB or a B rating. We thus restrict the sample of loans to these two ratings and, for each institution-quarter, we construct two observations: one for the cov-lite share for BB-rated loans, and the other for the cov-lite share for B-rated loans. In Internet Appendix Table 1 (Table IA1), we show results for regressions of the cov-lite share on our event indicators as well as their interactions with an indicator for B-rated loans. Consistent with LISCC institutions having significant regulatory concerns, these institutions increase the cov-lite share by about 14 percentage points more after the leveraged lending guidance for B-rated loans as compared to BB-rated loans. By contrast, for non-LISCCs, we do not find evidence of such a difference between B-rated loans and BB-rated loans while the guidance is in effect.

Overall, the results in this section are consistent with the guidance motivating more strictly regulated banks to originate a greater fraction of their syndicated loans as cov-lites, especially for those loans that are more likely to attract scrutiny if they remain on the balance sheet. These effects appear to begin as soon as the guidance is proposed, consistent with anticipation effects.

5.2. Stress test results

While the difference-in-differences results following the leveraged lending guidance are instructive, one potential weakness of this analysis is that, due to the time series nature of the guidance events, the

differences we find between LISCCs and non-LISCCs could be due to a differential impact of other macroeconomic events. In this subsection, we address this issue by considering shocks to individual institutions that do not apply to other institutions and occur at different points in time for different institutions. We obtain these shocks from the Federal Reserve Bank's annual CCAR Reports. These reports are available for the years 2011-2021, although individual banks' results were not disclosed prior to 2013.

In the CCAR process, the Federal Reserve evaluates whether US bank holding companies, savings and loan holding companies, and intermediate holding companies of foreign banks with \$100 billion (\$50 billion prior to 2018) or more in assets have adequate capital, whether they are able to absorb losses under several stress test scenarios, and whether their planned capital distribution policies are appropriate, along with other items of supervisory interest. Several months before the CCAR Report is generated, the Federal Reserve publishes the adverse scenarios that banks must use for their stress test modeling. Upon reviewing banks' results and capital plans, the Federal Reserve can either not object to a bank's capital plans, it can issue a conditional non-objection requiring the bank to resubmit its capital plan by a certain deadline to address identified weaknesses, but allowing planned capital distributions to occur until the resubmission, or it can object to the bank's capital plans, barring the bank from making capital distributions unless and until it resubmits a non-objectionable capital plan.

Both a conditional non-objection and an objection therefore indicate weaknesses in banks' capital plans. While a bank cannot adjust its capital position for a period that has already passed, one way the bank can enhance its ability to meet CCAR requirements for the next reporting period is to originate risky loans as cov-lite loans to make them easier to offload so the bank can show a better capital position if required for that period. Another way to put this is that, for a bank that may have difficulties meeting capital requirements, issuing cov-lite loans instead of cov-heavy loans makes it more likely that loans will not stay on the balance sheet and increases the bank's flexibility. Perhaps as a result of learning, most banks are subject to a CCAR objection or conditional non-objection just once (among banks active in the leveraged loan market this includes Ally Financial, BB&T, Bank of America, Capital One, Citigroup, Credit Suisse, HSBC North America Holdings, JPMorgan Chase, and RBS Citizen). Two banks failed to unconditionally

pass CCAR twice in non-consecutive years (Goldman Sachs and Morgan Stanley). One bank incurred a CCAR objection three years in a row (Deutsche Bank).

In Table 3, we show results of tests of whether a bank that does not unconditionally pass CCAR subsequently increases cov-lite origination as a fraction of total leveraged loan amounts originated. We control for launch quarter fixed effects to purge the general time trend in cov-lite issuance. We present results with and without lead agent fixed effects to address certain lead agents' greater propensity to originate cov-lites (e.g., due to specialization, reputation effects, etc.).

In Panel A of Table 3, we assess lending behavior in a one-year event window around the objection (or conditional non-objection). That is, we include an indicator variable that equals one during the four quarters prior to the objection to test for a pre-trend in behavior changes, and we include indicator variables that turn on between the CCAR report that flags the objection and the subsequent CCAR report (typically covering a period of four quarters, except for 2015/2016 when the time elapsed between the two reports is five quarters). In the case of consecutive objections, the pre-objection behavior is measured before the first objection, and the post-objection behavior is measured for as long as the objections continue. Odd numbered columns differentiate between conditional non-objections and outright objections while even numbered columns combine the two. Note that there are 29 conditional non-objection quarters and 19 objection quarters. Finally, we present results both for the full sample of financial institutions as well as a subsample restricted to those banks that are subject to CCAR stress testing.

As seen in Panel A, banks that fail to unconditionally pass CCAR do not exhibit cov-lite issuance behavior that is different from other banks in the year prior to this failure. Behavior changes appear to start within a year after the CCAR report. When not controlling for lead agent fixed effects, we find a roughly 16 percentage point increase in the cov-lite share for both objection types during the year after the CCAR objection/conditional non-objection. Controlling for lead agent fixed effects attenuates the results, with a significant increase of 6-8 percentage points seen for conditional non-objections while the result for objections is not significant. Combining the two types, we do not see a result that differs from zero statistically. However, Wald tests show that the insignificantly negative coefficient for the pre-objection

indicator and the insignificantly positive coefficient for the any-objection indicator are statistically significantly different from one another.

While using a one-year event window around the CCAR report helps pin down the timing of the lending response, this approach implicitly assumes that banks learn to originate cov-lites to enhance flexibility in their capital position for one year and then forget about the benefits of doing so and return to baseline. We address this issue in Panel B of Table 3, where we test for a permanent difference in lending behavior after the initial negative CCAR report. This change in specification results in larger coefficients particularly for full CCAR objections. In our cleanest specification, considering only CCAR banks and including lead agent fixed effects, the cov-lite share increases by a statistically significant 16 percentage points (Column (8)), a coefficient that is roughly three times larger than the coefficient for the one-year event window. Thus, the results appear to be consistent with a permanent learning effect, in which banks that have failed to pass a stress test outright rely more on cov-lite lending subsequently to increase balance sheet flexibility.

One possible alternative interpretation is that cov-lite lending may reflect risky underwriting practices that are correlated with risky capital positions that in turn attract CCAR scrutiny. Under this interpretation, cov-lite originations may increase over time due to macroeconomic effects, and the increase is taken up primarily by risk-loving banks and the CCAR indicator variables capture this differential time trend. However, this interpretation is inconsistent with our finding that there is no difference between banks prior to a CCAR objection, and it cannot explain why the change in lending response occurs immediately after the objection, especially since objection timing differs for different banks. We also note that Badoer, Emin, and James (2023) find that cov-lite loans are not riskier than cov-heavy loans ex post.

Overall, while the leveraged lending guidance analysis has some vulnerabilities to macroeconomic trends and the CCAR analysis has limitations due to just 12 banks active in the leveraged loan market failing to pass CCAR outright, the limitations of these two analyses seem orthogonal to each other and the results from both analyses are consistent with regulatory pressures as a likely contributor to banks' motivation to originate more cov-lite loans.

As a final test of the regulatory pressure explanation for the cov-lite trend, we use stress test results from the DFAST exercise to assess how exposed each sample bank is to what DFAST labels the supervisory severely adverse scenario. To that end, we subtract from the bank's actual capitalization ratio as of the DFAST exercise quarter the projection period minimum capital ratio, i.e., actual capitalization minus worst case capitalization. We refer to this variable as the bank's capitalization vulnerability. In Table 4, we regress the cov-lite share for each lead agent on capitalization vulnerability, launch quarter fixed effects and, in even numbered columns, lead agent fixed effects. The results in Table 4 show that at a given point in time banks with a more vulnerable capital position originate more cov-lites, consistent with a regulation-induced desire to be able to offload loans. These results are strongly statistically significant at the 1% level across banks, but the coefficient size shrinks by more than half within banks, with only two of five ratios showing statistical significance at the 10% level, possibly due to stickiness in the capitalization variables or the aforementioned learning effects, where a bank that has started to originate more cov-lites due to exposure to a regulatory shock does not stop originating cov-lites once the exposure is removed or alleviated.

6. The liquidity of leveraged loans

We now turn to an examination of the liquidity of cov-heavy and cov-lite loans. Such a comparison has to control for differences in loan characteristics. We therefore start with a comparison of loan characteristics between cov-heavy and cov-lite loans and between public and private cov-lite loans. We then present our liquidity metrics. As discussed earlier, our conjecture for the increase in issuance of cov-lite loans relative to cov-heavy loans relies on cov-lite loans being more liquid. Cov-heavy loans have adverse selection costs as they have to be monitored. We expect the benefit in reducing adverse selection costs to be higher for cov-lite loans to private firms than to public firms because adverse selection costs are less for public firms given their richer information environment. We conclude the section by testing these hypotheses. The remainder of the section introduces our liquidity metrics and tests our hypotheses.

6.1. Cov-lite loans versus cov-heavy loans

Panel A of Table IA2 compares characteristics of cov-lite loans versus cov-heavy loans. Hotchkiss and Jostova (2017) find that issue size is the most important determinant of trading volume for bonds. In our sample, deal size is much larger for cov-lite loans. The average and median deal sizes for cov-lite loans are roughly one and a half time the respective deal sizes for cov-heavy loans. Cov-lite loans also have a longer maturity.

When it comes to stocks, volatility and trading go together. Greater volatility means that more information is coming to the market about a stock, but that information is incorporated into prices through trading. If loans behave like stocks, we expect riskier loans to trade more. Hotchkiss and Jostova (2017) use a sample of bond prices from insurance companies to show that trading falls with risk for high yield bonds. In this case, we would expect riskier loans to trade less. Cov-lite loans are more likely to be rated than cov-heavy bonds. However, the ratings of cov-lite loans are lower than the ratings of cov-heavy loans by one notch at the median. More cov-lite loans are sponsored, meaning generally that the borrower has the support of a private equity firm or some equivalent entity. We compute proxies for sponsor reputation following Demiroglu and James (2010) and Shive and Forster (2022). We find that these proxies are higher for cov-lite loans than for cov-heavy loans.

As discussed in the previous sections, many cov-lite loans involve a package of tranches that includes a revolver tranche or a term loan A tranche. These two types of tranches are rarely sold to institutional investors. We find that 52.2% of the cov-lite loans have a revolver tranche. In contrast, 73.4% of cov-heavy loans have a revolver tranche. In addition, cov-heavy loans are almost ten times more likely to have a term loan A tranche than cov-lite loans (2.4% versus 22.4%). It follows from this that more than 45% of cov-lite loans have no tranche that involves bank monitoring. We expect the loans without bank monitoring to be riskier if there are no mitigating factors. As discussed before, one mitigating factor is often that the loan has a sponsor. We would also expect second lien facilities to be riskier. More cov-lite loans involve a second

lien facility than cov-heavy loans (18.0% versus 12.5%). Less than one tenth of the loans, whether cov-lite or cov-heavy, are asset-based loans. We expect loans to smaller firms to be less liquid.

Empirical evidence for bonds indicates that bonds of public firms trade more than bonds of private firms (Hotchkiss and Jostova, 2017). The typical cov-lite loan is to a private firm while the typical cov-heavy loan is to a public firm. A cov-lite loan is roughly 50% more likely to be a loan to a private firm than to a public firm. Specifically, 60.6% of cov-lite loans are to private firms. In contrast, 43.9% of cov-heavy loans are to private firms. However, these sample averages hide the trends we discussed in Section 4. In that section, we show that at the end of the sample period, highly leveraged loans to private firms are almost all cov-lite loans.

We have access to Compustat data for registered firms. We show that cov-lite borrowers have more assets, higher leverage, a higher current ratio, a lower coverage ratio, and a lower ratio of tangible assets, than cov-heavy borrowers.

It should be clear from these univariate comparisons that along many dimensions cov-lite loans are riskier than cov-heavy loans. The major exception is one that is known from the literature (Badoer, Emin, and James, 2024) and that we will discuss later in greater detail, namely that cov-lite loans are much more likely to be sponsored than cov-heavy loans.

6.2. Private versus public cov-lite borrowers

In Panel B of Table IA2, we compare cov-lite loans by registered borrowers (public firms) and non-registered borrowers (private firm).⁸ Private firm cov-lite loans are smaller than public firm cov-lite loans. The size difference is large as the mean of private firm cov-lite loans is \$699 million while the mean of public cov-lite loans is \$1,231 million. There is no significant difference in maturity. Almost all cov-lite loans are secured and rated irrespective of whether the borrower is private or public. Private firm cov-lite

⁸ Note that it is possible for a non-listed firm to be a registered borrower. For instance, it could be a formerly public firm that has public bonds outstanding from before it went private.

loans have a lower rating than public firm cov-lite loans by one notch at the median. There is an enormous difference in the extent of sponsorship as 90% of private-firm cov-lite loans are sponsored but only 48% of public-firm loans are sponsored. The differences in sponsor reputation between the two types of loans are significant but economically small. More private-firm cov-lite loans are part of an LBO than public-firm cov-lite loans (28.0% compared to 13.6%). More loans to private firms have revolvers than to public firms (53.7% versus 47.1%). Almost no loan to a private firm has a term loan A tranche, but 5.4% of the loans to public firms have such a tranche. More loans to private firms have a second lien facility. Finally, private firm borrowers are younger than public firm borrowers.

6.3. Liquidity metrics and hypotheses tests

We first present our liquidity metrics and show how they differ between cov-lite loans and cov-heavy loans. We deem a loan to be traded in the secondary market if LCD records a break price for the loan. Our data for the initial number of loan participation funds, for the average percentage of principal traded and for the percentage of months with a trade comes from the CRSP Mutual Fund database and captures the trading activity of loan participation funds. We have quote data from Thomson Reuters/LSTA Mark-to-Market pricing service.

Table 5 compares liquidity metrics for cov-lite loans to cov-heavy loans. We find that cov-lite loans are almost twice as likely to trade on secondary markets than cov-heavy loans. Specifically, 83% of cov-lite loans trade on secondary markets while 47% of cov-heavy loans do the same. The time-to-break for a loan is shorter for cov-lite loans. We also find that the initial number of loan participation funds is 5 on average for cov-lite loans but 2 on average for cov-heavy loans. The average percent of principal traded is twice as high for cov-lite as for cov-heavy loans – 1.3% versus 0.6%. We then report 3-month bid-ask spread averages. The bid-ask spread for the first three months and the first year, but not the second and third years, is significantly lower for cov-lite loans than for cov-heavy loans. We show later that the difference in the second and third years is significant once we account for the difficulty in renegotiating

distressed cov-lite loans (see Table 9). Finally, cov-lite loans that trade receive significantly more quotes than cov-heavy loans. All this evidence shows that cov-lite loans are more liquid than cov-heavy loans as expected with the cov-lite liquidity advantage hypothesis.

We now investigate the probability that a highly leveraged loan will trade given the characteristics of the loan. For this investigation, we estimate probit regressions for whether the loan will have a break price, which means whether it will start trading. We present average marginal effects for the probit regression. We control for loan and firm characteristics. We also use rating, year, and industry fixed effects. In Column (1) of Table 6, we estimate the regression for all loans. We use an indicator variable for cov-lite loans, so that the coefficient on that indicator variable estimates the average marginal effect when a loan is cov-lite as opposed to cov-heavy. We report that a cov-lite loan is more likely to trade than a cov-heavy loan. We therefore find support for the hypothesis that cov-lite loans are more liquid. We further report that larger loans are more likely to trade as expected. Loans with a longer maturity and sponsored loans are more likely to trade, while loans with a term loan A tranche are less likely to trade. In Column (2), we test the hypothesis that cov-lite loans from private firm borrowers have a bigger liquidity advantage than cov-lite loans from public borrowers. We add to the cov-lite loan indicator of Column (1) an indicator variable for public firms and a variable that is the interaction of the cov-lite loan indicator and the indicator variable for public firms. If cov-lite loans of private borrowers have a greater liquidity advantage, the interaction would have a negative coefficient. Here, it is not significant, so that the coefficient estimate is not evidence of a greater liquidity advantage of cov-lite loans of private borrowers relative to cov-lite loans of public borrowers. In Column (3), we re-estimate the regression in Column (1) but for rated firms only. We again find a significant positive coefficient for cov-lite loans. Finally, in Columns (4) and (5) we restrict the sample to public borrowers. In Column (4), we estimate the same regression as in Column (1). In Column (5), we add control variables from Compustat. We find that adding these variables has only a trivial impact on the coefficient on the cov-lite indicator variable. It follows from this discussion that cov-lite loans are more likely to trade, but whether the borrower is private or public does not seem to affect the probability that a highly leveraged loan trades.

Because we include private firms in our sample, we are unable to include firm financial control variables in our full sample regressions. We therefore rely on the assumption that the first lien rating adequately controls for risk for our sample loans. In Internet Appendix Table IA3, we test this assumption by estimating regressions predicting borrower bankruptcy subsequent to loan origination. We measure bankruptcy over a time horizon of up to five years using bankruptcy filing dates from Capital IQ. The global bankruptcy rate for our sample is 1% over a three-year horizon, and 3% over a five-year horizon. As shown in Table IA3, there is no difference in bankruptcy rates between cov-lite and cov-heavy loans once we control for the rating. It is also worth noting that, consistent with our arguments in Section 2, sponsored loans are less likely to be followed by bankruptcy than unsponsored loans.

We now turn to how quickly a highly leveraged loan trades. The regressions are the same as in Table 6 except that they are ordinary least squares regressions. In Column (1) of Table 7, we see that a cov-lite loan trades more quickly than a cov-heavy loan. In Column (2), contrary to Table 6, the indicator variable for public firms is negative and significant, indicating that highly leveraged loans with public borrowers trade more quickly. However, the interaction of registered firm and cov-lite is positive and mostly offsets the faster trading time of registered firms, so that the faster trading time of leveraged loans for public borrowers only holds for cov-heavy loans. Looking within firm type, cov-lite loans trade about 3.6 days faster than cov-heavy loans for private firms (compared to a sample average of 14 days), but cov-lite loans of public firms trade only about 1.9 days faster than public firm cov-heavies. Hence, here, the liquidity advantage of the cov-lite loans over the cov-heavy loans is greater for private borrowers than public borrowers.

A standard measure of liquidity is how much an asset trades. In Table 8, we estimate our regressions using the monthly average percentage of the loan amount traded by loan mutual funds, measured as the change in loan shares held as reported in CRSP. In Column (1) of Panel A, we use all loans and find that the percentage traded is higher for cov-lite loans. We also use the number of loan participation funds as an explanatory variable and find that the monthly average percentage traded increases with that number. We use the additional indicator variables in Column (2) that allow us to distinguish between loans with private borrowers and loans with public borrowers. We again find that the liquidity advantage of cov-lite loans is

greater for private borrowers than public borrowers. This relative advantage is measured by the interaction Covenant lite x Registered. The estimate for this coefficient is -0.377 and is significant at the 1% level. This relative advantage is large as it is almost of the same magnitude as the coefficient on the indicator variable Covenant lite loan which is 0.461. We show next that cov-lite loans trade more than cov-heavy loans if we restrict the sample to rated borrowers (Column (3)) and to public borrowers (Columns (4) and (5)). In Column (5) we use firm characteristics from Compustat in addition to our other regressors. Using these additional variables does not affect the coefficient on the Covenant lite indicator variable meaningfully. In Columns (6) to (10), we show regression estimates when the dependent variable is the percentage of months with trades. We find that cov-lite loans are more likely to trade in a given month. However, in Column (7), while the interaction has a negative coefficient, it is not significant.

One potential issue with trading volume reported by loan mutual funds is that one mutual fund may sell shares to another mutual fund, causing us to double count these shares when computing trading volume. In Columns (1) through (5) of Panel B, we remedy this issue by only considering the average percentage of the loan amount bought by loan mutual funds. Results are qualitatively similar to Panel B and remain highly statistically significant, although magnitudes are cut in half, consistent with loan mutual funds being more likely to trade cov-lite loans amongst themselves than cov-heavy loans. In Columns (6) through (10) of Panel B, we investigate the percentage of months with loan purchases. Again, we find a greater propensity of cov-lites to trade, and this time the interaction between Registered and Cov-lite is statistically significant at the 5% level in Column (7).

As discussed earlier, we have access to bid-ask spreads from Thomson Reuters/LSTA Mark-to-Market pricing service. In Panel A of Table 9, we estimate regressions using the bid-ask spread as the dependent variable. We use a sample of all loans and a sample of non-distressed loans. We allow for changes as the loan becomes older by estimating our regressions on subsamples that differ by the age of the loan. For both samples, we estimate regressions for the average bid-ask spread over the first three months of a loan, the first year, the second year, and the third year. For each subsample, we estimate the regression with the interaction between Cov-lite and Registered. We find for all Columns that cov-lite loans have narrower bid-

ask spreads. This difference is unequivocally significant for the first three months as well as the first year of the loan. For years 2 and 3, the difference between cov-lite and cov-heavy is not significant when we include loans priced as distressed, but is significant when we exclude them. Note that cov-lite loans do not offer a renegotiation trigger when firm performance declines and thus are mechanically more likely to be priced as distressed on the secondary market than cov-heavy loans, whose interest rate can be adjusted upwards if the deterioration in financial performance triggers a covenant violation. The interaction between Cov-lite and Registered is significant and positive in Columns (1) and (2) but not significant in Columns (3) and (4). In Columns (5) through (8), when we focus on non-distressed loans, the interaction is positive throughout, though significant only for Columns (5) and (6).⁹ This evidence is supportive of the hypothesis that cov-lite loans have a greater liquidity advantage for private borrowers.

In Panel B of Table 9, we use the number of quotes as the dependent variable. Loans that receive more quotes are more liquid in that there is potentially more competition for orders. We find that cov-lite loans have more quotes than cov-heavy loans, and this difference is significant for all time horizons. We also find that the cov-lite loans of private borrowers have a significantly greater liquidity advantage up to one year after origination. The coefficient on the interaction is slightly negative for the second and third years as well, but is not significant. An issue is that the sample shrinks as loans drop off the sample, so that the number of loans for which we have data in the third year is substantially smaller than the number of loans for which we have data in the first three months.

⁹ By market convention, loans are considered priced as distressed if the price is below 90 cents per dollar of principal. The interaction in Columns (5) and (6) is significant if we additionally drop loans priced below 95 cents. For a 5 to 6-year term loan, a price of 95 in year 3 corresponds to a yield-to-maturity that is about 150-200bp larger than the original yield of the loan, still indicating a sizeable deterioration in fundamentals that could have triggered a covenant violation and thus renegotiation for a cov-heavy loan.

7. The yields of highly leveraged loans

As discussed in the introduction and Section 2, for a given level of risk, a cov-lite loan should have a lower yield than a cov-heavy loan and the yield advantage of cov-lite loans should be greater for private borrowers than for public borrowers. We test these hypotheses in this section.

Yield comparisons are intrinsically difficult because what we would like to do is compare two loans, one cov-lite and one cov-heavy, that are the same except for the covenant differences. In general, one cannot perform comparisons that are close to the desired comparison. While some firms originate both types of loans in close succession, the choice to do so is clearly endogenous. We address this issue with an approach similar to Ivashina and Sun (2011) and Nadauld and Weisbach (2012), who analyze the within-deal difference in the credit spread between different tranches. Specifically, in Table 10, we regress the credit spread of the institutional tranche on the cov-lite indicator, the registered indicator, the interaction of the two, and, importantly, the credit spread on the revolving line of credit contained within the same deal.¹⁰ Thus, this sample is limited to loans that have both an institutional tranche and a revolver. Since both the institutional tranche and the bank tranche are issued at the same time and have the same priority, the pricing of the bank tranche can be expected to reflect any omitted variables related to the riskiness of the firm's cash flows and/or the value of its collateral. Controlling for the credit spread paid on the bank tranche should thus purge the effect of such omitted firm fundamentals. Following the above cited papers, we also restrict the sample to loans rated between BB+ and B- since 93% of cov-lite loans have a rating within this range. We include rating, industry, and year-quarter fixed effects.

In Column (1) of Table 10, we report estimates where we only include the revolver spread as a control variable. We find that after controlling for credit risk using the revolver spread, cov-lite loans are 36 basis points cheaper than cov-heavy loans. In Column (2), we add controls having to do with the loan or the firm. However, since we use loans from private borrowers as well as public borrowers, we do not have access to

¹⁰ As discussed earlier, a tranche A is rare for private firms, so that we cannot use the tranche A credit spread as benchmark for our analysis.

Compustat for the whole sample. Further, since we are comparing the yield of an institutional tranche to a revolver, we also include characteristics of the revolver. The indicator for a cov-lite loan remains negative and significant at 15 basis points. We find that the yield increases with the revolver spread, increases if the loan is asset based, and falls with the amount and maturity of the institutional tranche. The yield also falls with the existence of a second lien facility, which enhances the first lien's collateral protection by adding a subordinated layer in the priority structure. In Column (3), we add an interaction between the cov-lite indicator and an indicator for public firms. We find that this indicator is positive and significant. It follows that cov-lite loans by private borrowers have a greater yield advantage over cov-heavy loans than cov-lite loans by public borrowers do. In the last two columns, we estimate the regression for the sample of public firms. Consistent with column (3), for that subsample, we find that the coefficient for cov-lite loans is numerically smaller and not statistically significant. For the regression in Column (5), we add Compustat variables as we did in the liquidity regressions that use only public firms. No Compustat variable is significant, consistent with the spread paid on the revolver capturing firm fundamentals.

Our results thus far establish that cov-lite loans are more liquid than cov-heavy loans and that, controlling for credit risk contained in ratings and revolver spreads, cov-lite loans pay lower yields. We hypothesize that these lower yields reflect that liquidity is priced in the secondary loan market. In Table 11, we test that hypothesis by regressing the credit spread paid on the first lien institutional tranche on the cov-lite indicators, the bid-ask spread, the number of loan participation funds that initially hold the loan along with our other control variables. The sample in this regression is restricted to traded loans originated from 2010 through 2021 so that bid-ask spread data are available. As in Table 10, we require the loans to have a rating between BB+ and B-. We further require that the traded tranche be a first lien institutional tranche. In Columns (1) and (2), we reproduce the results from Columns (3) and (5) of Table 11 for this smaller sample, except we add the number of loan participation funds involved. The coefficients for the cov-lite indicator and its interaction with the registered indicator remain numerically similar, although having dropped more than 50% of sample observations we lack the power to detect a statistically different effect

for private vs. public firms and to detect statistical significance for public firms alone. Nevertheless, the cov-lite indicator for the full sample remains significant at the 1% level.¹¹

In Columns (3) and (4), we add the loan's average quoted bid-ask spread over the three months post-origination as a "predictor" of the credit spread. Note that the quoted bid-ask spread can only be observed after the credit spread has been set and the loan starts trading. As such, the underlying assumption in these regressions is that the ex-post observed bid-ask spread is highly correlated with the ex-ante expected bid-ask spread, and the regressions aim to test whether the expected bid-ask spread is priced. Columns (3) and (4) show that irrespective of adding or not adding Compustat variables to the regression, the loan's bid-ask spread is strongly priced. The coefficient implies that a 32 basis point increase in the bid-ask spread (one standard deviation) is associated with a 19 basis point increase in the institutional credit spread (0.32×58.815). In Columns (5) through (10), we use the average bid-ask spread over the first, second, and third year post-origination, respectively. Results are qualitatively similar to Columns (3) and (4), except that, as one might expect when using ex-post bid-ask spreads to measure expected liquidity at origination, the further away from origination bid-ask spreads are measured, the smaller their coefficient size.

One concern with Table 11, especially since we are measuring bid-ask spreads after measuring credit spreads, is that bid-ask spreads could be subject to endogeneity and/or reverse causality. We address this issue using an instrumental variables regression. To that end, we instrument a loan's bid-ask spread using the industry average bid-ask spread on the trading day preceding the loan origination date. Industry definitions are from LCD, and there are 31 different industries in our full sample. We thus rely on the industry-level bid-ask spread predicting the loan's bid-ask spread, but, conditional on control variables, not being correlated with the loan's credit spread other than through the bid-ask spread.

An obvious concern is that different industries are subject to different variation in creditworthiness over time so that the loan's credit spread could be linked to industry-level variation in credit spreads that in turn is reflected in the industry-level bid-ask spread. Our year-quarter fixed effects will not pick up this variation

¹¹ In unreported regressions, we confirm that the change in statistical significance level is not driven by inclusion of the loan participation fund variable.

if it differs across industries. At the same time, because our instrument is at the industry-time level, we cannot include industry x year-quarter fixed effects in the estimation. To address this concern, we therefore augment the regression in two ways. First, we interact the rating fixed effects with the year-quarter fixed effects. So long as the industry-level variation in creditworthiness is reflected in issuers' first lien ratings, this interaction will cure the aforementioned concern. Although one may worry that rating agencies may be slow to update ratings over the life of a loan, we note that we study newly originated loans, and issuance of a new loan is much more likely to trigger a review of the firm's first lien rating than other events not directly related to new debt issuance. Secondly, we include the industry average revolver spread over the quarter prior to loan origination as an additional control for industry-level creditworthiness to the regressions.¹² One way to interpret this additional control variable is that if it does not load, then the relevant industry-level time series variation in creditworthiness is already captured by the rating x year-quarter fixed effects.

Table 12 shows the first and second stages of the instrumental variables estimation. We limit the estimation to bid-ask spreads over the first three months or the first year of the loan's existence since longer horizon bid-ask spreads are less correlated with origination-time liquidity expectations and because unreported regressions show that our instrument is not significantly correlated with bid-ask spreads measured that far away from the measurement of the instrument. Column (1) of Table 12 reports results from the first stage predicting the 3-month loan-level average bid-ask spread using the pre-origination industry average bid-ask spread. After controlling for all other variables, the industry average bid-ask spread significantly predicts the loan's 3-month bid-ask spread. The coefficient is an economically modest 0.035 basis points increase in spread for each one basis point increase in the industry-level spread. Column (3) reports the second stage, using the instrumented 3-month bid-ask spread to predict the institutional credit spread. We find that a 32 basis point increase in the instrumented bid-ask spread results in a 157 basis point

¹² To maximize statistical power, unlike in Table 11, we do not restrict the sample for instrumental variables estimation to loans that are accompanied by a revolving line of credit. However, in Internet Appendix Table IA4, we show that when we substitute the spread on the deal's own revolver, whenever available, for the industry-level revolver spread, our results discussed hereafter remain highly similar quantitatively and qualitatively.

increase in the institutional credit spread (0.32×491), statistically significant at the 5% level. Thus, the effect is eight times as large as in the OLS regression.

The instrument weakness test at the bottom of the table provides a clue as to why the coefficient is so much larger. The Kleibergen-Paap rk Wald F-statistic is 8.820. While critical values for this specific test statistic have not been estimated in the literature, Stock and Yogo (2005) provide critical values for the related Cragg-Donald Wald statistic which assumes i.i.d. errors. We reproduce their critical values for a single endogenous regressor and one instrument in Table 12. For example, to ensure that the true size of a 5% significance test does not exceed 10%, the F-statistic should be at least 16.38. For a 15% maximal size, the F-statistic should exceed 8.96. We conclude that the second stage estimation alone does not allow us to guarantee that the true significance level of the bid-ask spread in this regression is better than 15% and our point estimate for the coefficient may be unreliable.

However, the econometrics literature does provide ways to test for statistical significance in the presence of weak instruments. In an IV estimation with one endogenous regressor and one instrument, both the Anderson-Rubin (1949) Wald test as well as the related Stock-Wright (2000) LM S statistic provide a test of the null that the endogenous regressor in the second stage is zero. Unlike the second stage t-statistic, these χ^2 -distributed test statistics are unaffected by instrument weakness. Column (3) of Table 12 shows that both these test statistics are at about 6.8, which is statistically significant at the 1% level. Thus, we can state with at least 99% confidence that the 3-month bid-ask spread is priced in the institutional spread even though we cannot provide a reliable point estimate for the IV. In addition, we note that the industry average revolver spread is insignificant in both the first and the second stage, which strongly suggests that the year-quarter $\times$ rating fixed effects fully capture relevant industry-level time series variation in credit risk, thus eliminating our primary concern about instrument validity.

Columns (2) and (4) of Table 12 repeat the analysis for the one-year bid-ask spreads. Results are similar. The industry average bid-ask spread is a slightly better predictor of one-year bid-ask spreads than of 3-month bid-ask spreads resulting in a higher first stage t-statistic and a higher Kleibergen-Paap rk Wald F statistic of 13.317. However, the instrument remains weak and we cannot rule out that a 5% significance

test has a true size between the 10% and the 15% significance level. The second stage coefficient is one third smaller than for the 3-month bid-ask spread and barely misses 5% estimated significance at a t-statistic of 1.94. Nonetheless, the Anderson-Rubin (1949) and Stock-Wright (2000) test statistics continue to reject the null hypothesis at the 5% level.

8. Conclusion

In this paper, we advance the hypothesis that the dramatic increase in the use of cov-lite loans in the leveraged loan market is in part the product of the regulators' attempt to limit banks' issuance of leveraged loans. Increased attention by regulators to leveraged loans made it potentially expensive for banks to hold such loans on their balance sheet. As a result, it became more valuable for banks to issue loans expected to be more liquid, so that they could sell them more easily. Cov-lite loans are more liquid than cov-heavy loans, so that if banks put a premium on loans that could be sold more easily, they would have issued more cov-lite loans. We show in this paper that the cov-lite share of leveraged loan issuances of the most strictly regulated banks increased relative to the share of other institutions after regulators introduced guidance on leveraged loans and that banks that fail to pass a stress test without reservations subsequently increase the cov-lite share of leveraged loan issuances as well. These results are surprising and would appear to be an unintended consequence of the greater regulatory attention to leveraged loans because everything else equal cov-heavy loans are safer. A further surprising result is that the impact of the increased regulatory attention is greater for issuance of loans by private borrowers than public borrowers. Our explanation for this result is that the liquidity advantage of cov-lite loans is greater for loans of private borrowers than public borrowers. We then demonstrate that cov-lite loans are more liquid than cov-heavy loans. We find that cov-lite loans are more likely to trade, have a shorter time-to-break, trade more, have tighter bid-ask spreads, and have more quotes. We show that the liquidity advantage of cov-lite loans over cov-heavy loans is greater for private issuers for all our liquidity metrics except for the likelihood of trading. Finally, we show

that the liquidity advantage of cov-lites is priced, so that cov-lites with the same rating as cov-heavyies have a lower yield.

Appendix A. Variable definitions

Variable	Definition	Source
<i>Deal characteristics (loans and bonds)</i>		
Deal size	Total deal amount combining all tranches of a deal, expressed in 2021 US dollars	LCD, Bureau of Labor Statistics
Maturity	Maturity of the debt expressed in years	LCD
Senior	Indicator equals one if the debt is senior, zero otherwise	LCD
Rating	Loans: First lien bank loan rating from S&P. Moody's rating substituted where not available. Corporate issuer rating substituted if no first lien rating available. Bonds: At-issue corporate issuer rating from S&P. Moody's rating substituted if unavailable. Fitch rating substituted if neither S&P nor Moody's available. Rating values: 0 indicates no rating. 1 indicates a rating of BBB- or better. 2, 3, 4, etc. indicate ratings of BB+, BB, BB-, etc. Ratings of CCC+ or worse have a value of 8.	LCD
Institutional spread	Credit spread paid on a loan's first lien institutional tranche	LCD
Revolver spread	Credit spread paid on a loan's revolving line of credit tranche	LCD
<i>Loan characteristics (not available for/applicable to bonds)</i>		
Rated	Indicator equals one if the issuer has a rating, zero otherwise	LCD
Traded on secondary market	Indicator equals one if the issue has a break price	LCD
Time-to-break	Days elapsed between break date and loan launch date	LCD
Break price	Price at which loan breaks into trading on the secondary loan market	LCD
Break below 99	Indicator equals one if the break price is less than 99 cents on the dollar, zero otherwise	LCD
Initial # LP Funds	Number of loan participation funds that report holding shares of a loan within one month of	CRSP Mutual Funds database

	the break date. Set to zero if no loan participation funds report holding shares.	
Average % principal traded	Average monthly percentage of the loan's principal traded by loan participation mutual funds.	CRSP Mutual Funds database
% months with trade	Percentage of months in which at least one loan participation mutual fund reports a change in the number of loan shares it holds	CRSP Mutual Funds database
Bid-ask spread	Average over a given period of the difference between the secondary market ask quote and bid quote for a loan divided by the ask quote and multiplied by 100	Thomson Reuters/LSTA Mark-to-Market pricing service
Number of quotes	Average over a given period of the number of brokers providing quotes for a loan in the secondary market	Thomson Reuters/LSTA Mark-to-Market pricing service
Sponsored	Indicator equals one if the loan is sponsored by a private equity firm. Includes initial LBO loans and refinancings.	LCD
LBO	Indicator equals one if the purpose of the loan is to finance a leveraged buyout. LBO loans are a subset of sponsored loans.	LCD
Revolver	Indicator equals one if the loan contains a revolving line of credit.	LCD
Term loan A	Indicator equals one if the loan contains a term loan A.	LCD
Second lien facility	Indicator equals one if the loan contains a second lien facility	LCD
Asset based loan	Indicator equals one if the loan is flagged as an asset based loan by LCD.	LCD

Firm characteristics

Unregistered firm	No asset data available for the fiscal year prior to debt issuance in either Compustat or Capital IQ. In the case of bond issuance, asset data also cannot be available three years prior to the last fiscal year since availability of such data is likely to be caused by the registration of the bonds themselves.	Compustat, Capital IQ
-------------------	---	-----------------------

Firm age	Year of debt issuance minus year the firm was founded. In the case of new legal entities created as a result of a spin-off, asset sale, or merger, the year the predecessor was founded is used.	Capital IQ, Google searches
Assets	Total book assets in \$ millions expressed in 2021 US dollars.	Compustat, Capital IQ, Bureau of Labor Statistics
Leverage	Long-term debt plus debt in current liabilities divided by total assets.	Compustat, Capital IQ
Current ratio	Current assets divided by current liabilities.	Compustat, Capital IQ
Coverage ratio	EBITDA divided by interest expense.	Compustat, Capital IQ
Tangibility	Net property, plant and equipment divided by total assets.	Compustat, Capital IQ

The following variables are winsorized at the 1st and 99th percentile: maturity, bid-ask spread, leverage, current ratio, $\text{Ln}(1 + \text{coverage ratio})$, and tangibility.

References

- Abuzov, Rustam, Christoph Herpfer, and Roberto Steri, 2025, Do banks compete on non-price terms? The dollar value of loan covenants, unpublished working paper.
- Ayotte, Kenneth, and Patrick Bolton, 2011, Covenant lite lending, liquidity, and standardization of financial contracts, in: Research Handbook on the Economics of Property Law, Kenneth Ayotte and Henry E. Smith (eds.), Edward Elgar, Cheltenham, UK.
- Badoer, Dominique C., Mustafa Emin, and Christopher M. James, 2024. Contracting costs, covenant-lite lending, and reputational capital. *Journal of Financial and Quantitative Analysis* 59, 3376-3415.
- Becker, Bo, and Victoria Ivashina, 2025, Covenant-light contracts and creditor coordination, unpublished working paper.
- Benmelech, Efraim, and Nittai K. Bergman, 2011. Bankruptcy and the collateral channel. *The Journal of Finance* 66, 337-378.
- Berlin, Mitchell, Gregory Nini, and Edison G. Yu, 2020, Concentration of control rights in leveraged loan syndicates, *Journal of Financial Economics* 137, 24-271.
- Billett, Matthew T., Redouane Elkamhi, Latchezar Popov, and Raunaq S. Pungaliya, 2016. Bank skin in the game and loan contract design: Evidence from covenant-lite loans, *Journal of Financial and Quantitative Analysis* 51, 839-873.
- Block, Joern, Jang, Young Soo, Kaplan, Steven N, and Schulze, Anna, 2024, A Survey of Private Debt Funds, *The Review of Corporate Finance Studies*, Volume 13, Issue 2, 335–383.
- Bradley, Michael, and Michael R. Roberts, 2015. The structure and pricing of corporate debt covenants, *The Quarterly Journal of Finance* 5.
- Bräuning, Falk, Victoria Ivashina, and Ali Ozdagli, 2025. High-yield debt covenants and their real effects. *The Review of Financial Studies*, forthcoming.
- Chava, Sudheer, and Michael Roberts, 2008. How does financing impact investment? The role of debt covenants, *Journal of Finance* 63, 2085 - 2121.
- Chernenko, Sergey, Ialenti Robert, and David S. Scharfstein, 2025, Bank Capital and the Growth of Private Credit, unpublished working paper.
- Dang, Tri Vi, Gary B. Gorton, Bengt R. Holmstrom, 2019, The information view of financial crises, *Annual Review of Financial Economics*, forthcoming.
- Diamond, Douglas W., Yunzhi Hu, and Raghuram G. Rajan, 2019, Pledgeability, industry liquidity, and financing cycles, *Journal of Finance* 75, 419-461.
- Duffie, Darrell. *Fragmenting Markets: Post-Crisis Bank Regulations and Financial Market Liquidity*. Walter de Gruyter GmbH & Co KG, 2022.
- Fleckenstein, Quirin, Manasa Gopal, German Gutierrez, and Sebastian Hillenbrand, 2020. Nonbank lending and credit cyclicity. Unpublished working paper.

- Gorton, Gary, and George Pennacchi, 1990, Financial Intermediaries and Liquidity Creation, *Journal of Finance* 45, pp. 49-71.
- Ivashina, Victoria, and David Scharfstein, 2010. Bank lending during the financial crisis of 2008. *Journal of Financial economics* 97, 319-338.
- Ivashina, Victoria, and Zheng Sun, 2011. Institutional demand pressure and the cost of corporate loans, *Journal of Financial Economics* 99, 500-522.
- Jang, Young Soo, 2025. Are Direct Lenders More Like Banks or Arm's-Length Investors? Unpublished working paper.
- Jiang, Sheila, Shohini Kundu, and Douglas Xu, 2022. Monitoring with small stakes: Evidence from leveraged loans. Unpublished working paper.
- Lee, Seung Jung, Dan Li, Ralf R. Meisenzahl, and Martin J. Sicilian, 2019, The U.S. syndicated term loan market: Who holds what and when? FEDS Notes (<https://www.federalreserve.gov/econres/notes/fedsnotes/the-us-syndicated-term-loan-market-20191125.html>, retrieved in December 2025).
- Matvos, Gregor, 2013. Estimating the benefits of contractual completeness, *The Review of Financial Studies* 26, 2798-2844.
- Nadauld, Taylor D., and Michael S. Weisbach, 2012, Did securitization affect the cost of corporate debt? *Journal of Financial Economics* 105, 332-352.
- Nelson, Bill, Greg Baer, and Jeremy Newell, 2017. Why leveraged lending guidance is far more important, and far more misguided, than advertised. Bank Policy Institute.
- Nini, Greg, David C. Smith, and Amir Sufi, 2012, Creditor control rights, corporate governance, and firm value, *Review of Financial Studies* 25, 1713-1761.
- Parlour, Christine, and Guillaume Plantin, 2008, Loan sales and relationship banking, *Journal of Finance* 63, 1291-1314.
- Roberts, Michael, 2015, The role of dynamic renegotiation and asymmetric information in financial contracting, *Journal of Financial Economics* 116, 61-81.
- Stock, James H. and Yogo, Motohiro, 2005, Testing for Weak Instruments in Linear IV Regression. *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, edited by Donald W.K. Andrews and James H. Stock, Cambridge: Cambridge University Press, chapter 5, pp. 80-108.

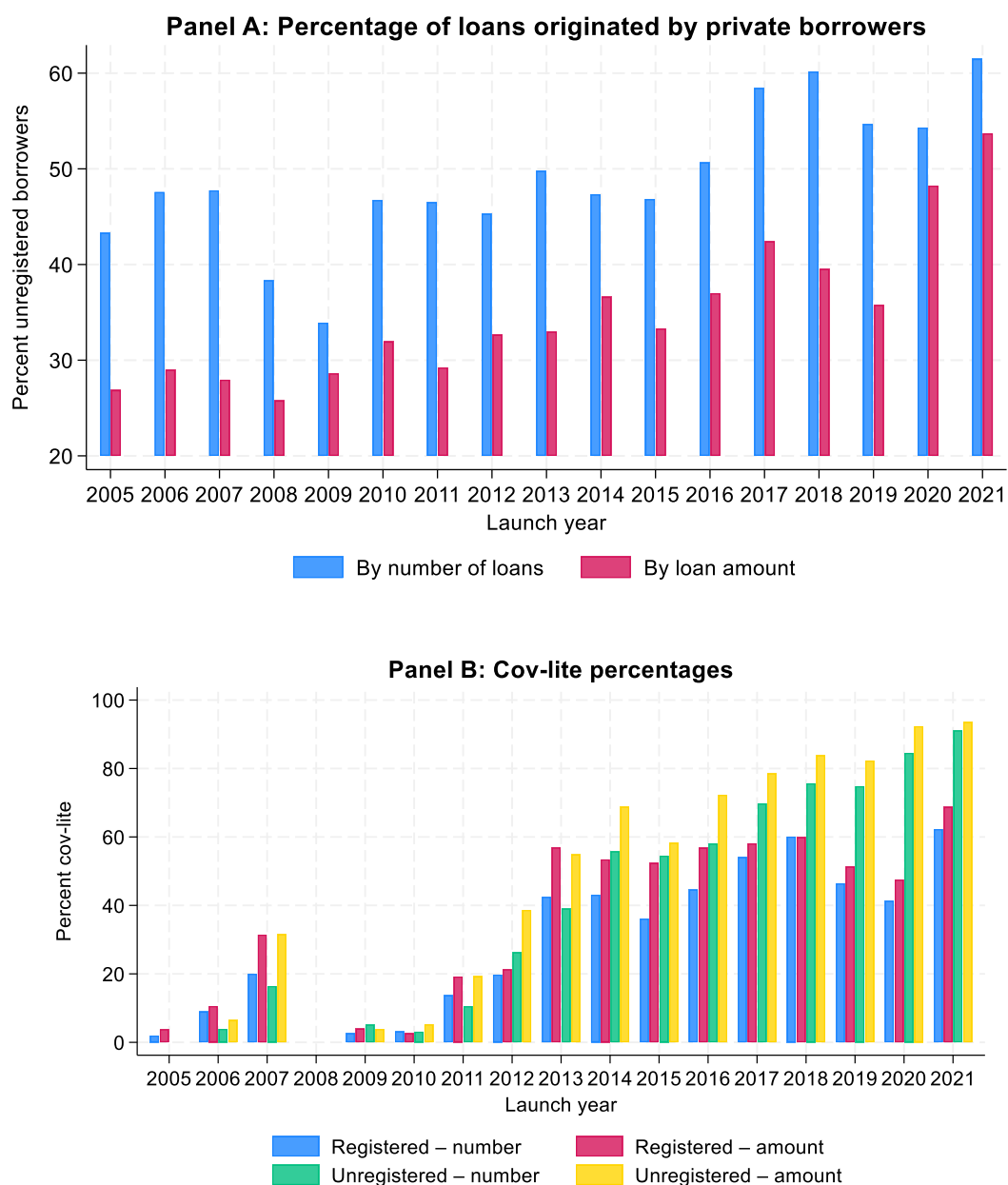


Figure 1: Loan market issuance trends over time

Panel A of this figure shows the percentage of loans (by number and loan amount) that are issued by borrowers not registered with the SEC. Panel B shows the percentage of the number as well as the amount of loans that are cov-lite for registered vs. unregistered issuers. The sample includes all loans issued by US issuers in the US leveraged loan market from 2010-2021 that meet our filter criteria. Specifically, it excludes loans to financial firms and utilities as well as add-on tranches to existing loans.

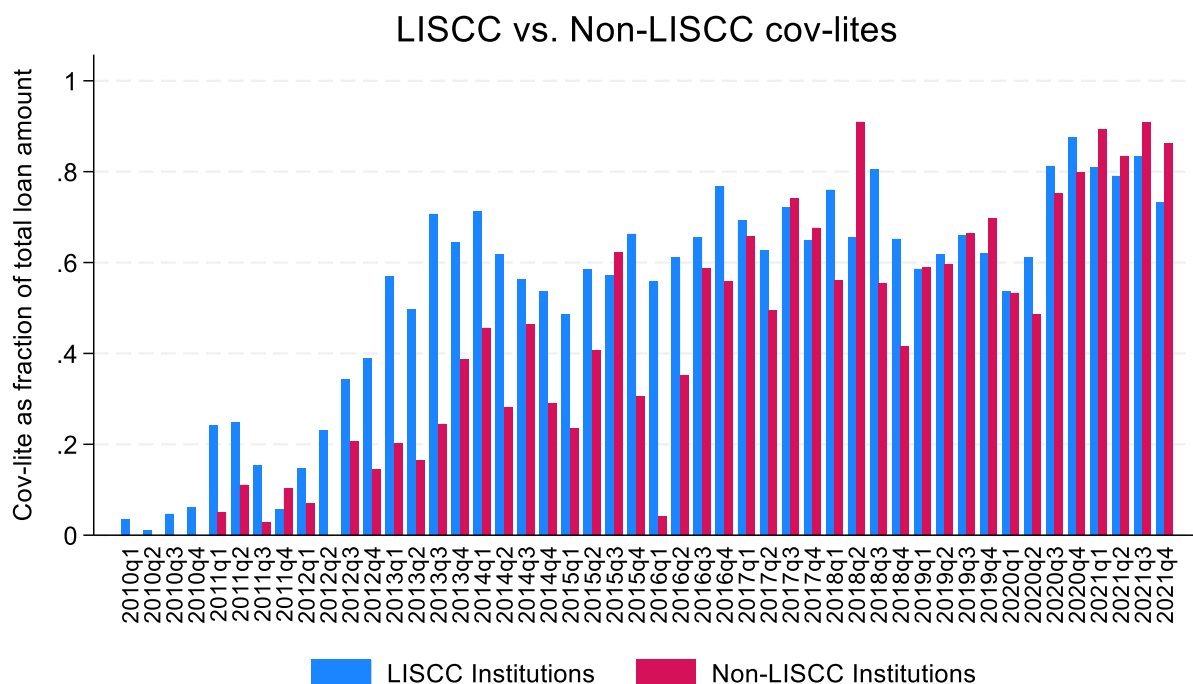


Figure 2: Covenant-lite loans as a fraction of total issuance for LISCC banks vs. non-LISCC institutions

This figure shows the fraction of total loan issuance by LISCC banks vs. non-LISCC financial institutions that is covenant-lite. The sample includes all loans issued by US issuers in the US leveraged loan market from 2010-2021. Total issuance volume during that time period by LISCC institutions is \$3,966 billion. Total issuance volume by non-LISCC institutions is \$760 billion. Non-LISCC institutions include domestic banks (\$314 billion), foreign banks (\$181 billion), and non-bank institutions (\$265 billion). Non-bank institutions include investment banks not affiliated with a bank holding company as well as asset managers.

Table 1: Data on Leveraged Issuance

This table shows issuance volume for covenant-heavy loans and covenant-lite loans originated by US issuers during the 2005-2021 sample period for the US leveraged loan market. The data source is Capital IQ Leveraged Commentary and Data (LCD).

	Covenant-heavy			Covenant-lite			Total	
	\$	N	% of total \$	\$	N	% of total \$	\$	N
2005	249.03	621	97.22%	7.12	7	2.78%	256.16	628
2006	343.33	656	90.62%	35.54	46	9.38%	378.87	702
2007	284.89	563	68.51%	130.92	126	31.49%	415.82	689
2008	114.88	266	100.00%	0.00	0	0.00%	114.88	266
2009	53.20	165	96.05%	2.19	6	3.95%	55.38	171
2010	188.90	400	96.53%	6.79	13	3.47%	195.69	413
2011	256.19	466	80.78%	60.96	65	19.22%	317.15	531
2012	273.86	493	73.02%	101.18	145	26.98%	375.04	638
2013	200.62	408	43.77%	257.74	281	56.23%	458.36	689
2014	157.94	343	40.93%	227.98	331	59.07%	385.92	674
2015	151.40	288	45.66%	180.15	233	54.34%	331.55	521
2016	143.72	256	37.45%	240.06	271	62.55%	383.78	527
2017	160.65	237	33.23%	322.82	408	66.77%	483.46	645
2018	142.45	194	30.61%	322.95	441	69.39%	465.39	635
2019	143.88	199	37.60%	238.81	324	62.40%	382.69	523
2020	92.59	151	30.85%	207.56	278	69.15%	300.15	429
2021	117.69	161	17.94%	538.32	644	82.06%	656.01	805

Table 2: LISCC institutions' response to leveraged lending guidance events

This table regresses the fraction of loan amounts each quarter that is cov-lite for each lending institution on time and institution type indicators and their interactions. The sample period includes the years 2010 (first year in which LISCCs are defined) through 2021. Since CCAR banks are known only from Quarter 4 of 2010 onwards, we set CCAR equal to one in early 2010 for those banks that are in CCAR in Quarter 4 of 2010. We update banks' CCAR status whenever the Federal Reserve Bank issues the scenarios for the next stress test with a list of CCAR banks that are subject to the test (Quarter 4 of the years 2010-2014, and Quarter 1 of the years 2016-2021). Beginning in 2019, some smaller CCAR banks are moved to a biannual testing cycle. Nonetheless, they know that they will continue to be subject to CCAR, and therefore are continuously classified as CCAR banks. *Previous year* refers to the period from Quarter 2 of 2011 through Quarter 1 of 2012. *Post proposal* covers the period after publication of the proposed guidance but before publication of the final guidance (Quarter 2 of 2012 through Quarter 1 of 2013). *Post guidance* refers to the period after publication of the final guidance but before publication of the clarification of the guidance (Quarter 2 of 2013 through Quarter 4 of 2014). *Post clarification* refers to the period after publication of the clarification but before the OCC statement on non-enforcement of the guidance (Quarter 1 of 2015 through Quarter 1 of 2018). *Post OCC statement* refers to the period after the OCC statement (Quarter 2 of 2018 onwards). *LISCC* is an indicator variable that equals one for all LISCC institutions, and *Non-LISCC CCAR* is an indicator that marks all CCAR banks that are not subject to LISCC supervision. Only one institution changes LISCC status during the sample period without leaving the sample: UBS AG. Columns (3) and (6) drop loans originated by investment banks or nonbanks to compare only commercial banks to each other. Standard errors are adjusted for clustering by lead agent and launch quarter. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	(1)	(2)	(3)	(4)	(5)	(6)
Previous year	0.046 (1.26)	0.061 (1.31)	0.081 (1.44)	0.039 (0.94)	0.048 (0.86)	0.096 (1.29)
Post proposal	0.070*** (2.79)	0.079** (2.47)	0.045 (1.37)	0.054*** (3.91)	0.062*** (2.86)	0.071*** (3.90)
Post guidance	0.138*** (3.29)	0.173** (2.66)	0.143** (2.18)	0.153*** (2.95)	0.187*** (2.80)	0.144** (2.15)
Post clarification	0.327*** (6.03)	0.405*** (6.24)	0.300*** (4.02)	0.298*** (4.78)	0.324*** (3.95)	0.263*** (3.41)

Post OCC statement	0.535*** (8.40)	0.524*** (6.03)	0.338*** (3.18)	0.472*** (6.54)	0.465*** (5.04)	0.355*** (4.85)
LISCC	0.053** (2.03)	0.052** (2.07)	0.062* (1.94)	-0.345*** (-3.82)	-0.245*** (-2.79)	-0.276*** (-2.99)
Previous year x LISCC	0.045 (0.79)	0.030 (0.49)	0.019 (0.27)	0.052 (0.82)	0.043 (0.56)	0.004 (0.04)
Post proposal x LISCC	0.283*** (4.91)	0.274*** (3.99)	0.335*** (4.89)	0.299*** (6.27)	0.291*** (5.61)	0.309*** (4.73)
Post guidance x LISCC	0.384*** (6.36)	0.349*** (4.72)	0.403*** (5.15)	0.368*** (5.55)	0.334*** (4.34)	0.401*** (5.03)
Post clarification x LISCC	0.249*** (3.37)	0.172** (2.10)	0.276*** (3.03)	0.262*** (3.21)	0.235** (2.33)	0.312*** (3.21)
Post OCC statement x LISCC	0.140 (1.67)	0.150 (1.44)	0.331*** (2.69)	0.182* (1.97)	0.189* (1.69)	0.315*** (3.09)
Non-LISCC CCAR		-0.004 (-1.16)	0.000 (0.00)		0.211** (2.14)	0.196* (1.97)
Previous year x Non-LISCC CCAR		-0.061 (-1.31)	-0.081 (-1.44)		-0.033 (-0.62)	-0.083 (-1.10)
Post proposal x Non-LISCC CCAR		-0.042 (-0.81)	-0.008 (-0.15)		-0.017 (-0.64)	-0.026 (-0.60)
Post guidance x Non-LISCC CCAR		-0.089 (-1.19)	-0.059 (-0.78)		-0.166 (-1.66)	-0.123 (-1.22)

Post clarification x Non-LISCC CCAR		-0.197** (-2.67)	-0.092 (-1.16)		-0.156 (-1.44)	-0.092 (-0.91)
Post OCC statement x Non-LISCC CCAR		0.025 (0.22)	0.211* (1.71)		-0.105 (-0.95)	0.006 (0.07)
Lead agent effects	No	No	No	Yes	Yes	Yes
Observations	1190	1190	1004	1190	1190	1004
R-squared	0.3448	0.3558	0.3971	0.5830	0.5865	0.5691

Table 3: Lender response to CCAR objections

This table regresses the fraction of loan amounts each quarter that is cov-lite for each lending institution on institution type and CCAR action indicators. The sample period includes the years 2010 (first year in which LISCCs are defined) through 2021. *Pre-objection* refers to the four quarters before the Federal Reserve Bank objects or conditionally does not object to the bank's capitalization plans in the CCAR report. Panel A assesses lending behavior during the year prior to the objection vs. the year after the objection. Panel B assesses whether an objection is associated with a permanent shift in lending behavior. *CCAR Conditional Non-objection* refers to the four quarters after the Federal Reserve Bank's CCAR report states non-objection to the bank's capitalization plans subject to certain conditions. *CCAR Objection* refers to the four quarters after the Federal Reserve Bank objects to the bank's capitalization plans in the CCAR report. *Any objection* refers to the four quarters after the Federal Reserve Bank either objects or conditionally does not object to the bank's capitalization plans in the CCAR report. *CCAR Conditional Non-objection (post)*, *CCAR Objection (post)*, and *Any objection (post)* refer to all quarters after the Federal Reserve Bank first conditionally non-objects/objects to the bank's capitalization plans. *LISCC* is an indicator variable that equals one for all LISCC institutions, and *CCAR* is an indicator variable that equals one for all CCAR institutions. Columns (1) through (4) cover the entire sample, while columns (5) through (8) are restricted to observations where the institution is a CCAR institution. In the full sample in Panel A, there are 41 pre-objection quarters, 29 conditional non-objection quarters and 19 objection quarters. The CCAR sample drops five pre-objection quarters due to the institution receiving a negative mention in its first CCAR report. In Panel B, there are 132 quarters post-conditional non-objection and 68 post-objection quarters. Standard errors are adjusted for clustering by lead agent and launch quarter. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

Panel A: Lending behavior in a one-year event window around the objection

	Full sample				CCAR banks only			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-objection	0.022 (0.42)	0.022 (0.42)	-0.061 (-1.25)	-0.059 (-1.27)	0.014 (0.23)	0.014 (0.23)	-0.084 (-1.65)	-0.082 (-1.66)
CCAR conditional Non-objection	0.167*** (2.89)		0.082** (2.27)		0.164** (2.37)		0.064* (1.71)	
CCAR objection	0.151*** (3.13)		-0.015 (-0.30)		0.166*** (2.96)		0.025 (0.47)	
Any objection		0.161***		0.049		0.165**		0.052

		(2.97)		(1.11)		(2.48)		(1.34)
CCAR	-0.078* (-1.77)	-0.078* (-1.77)	0.031 (0.74)	0.025 (0.62)				
LISCC	0.265*** (5.70)	0.265*** (5.70)	-0.018 (-0.42)	-0.025 (-0.58)	0.279*** (4.15)	0.279*** (4.16)	-0.007 (-0.66)	-0.011 (-0.99)
Lead agent effects	No	No	Yes	Yes	No	No	Yes	Yes
Launch quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald test:								
Any objection = pre-objection								
F-statistic (p-value)		6.67** (0.013)		5.22** (0.027)		5.04** (0.032)		4.93** (0.033)
Observations	1190	1190	1190	1190	607	607	607	607
R-squared	0.3896	0.3896	0.6024	0.6020	0.4557	0.4557	0.5876	0.5875

Panel B: Lending behavior post- vs. pre-objection

	Full sample				CCAR banks only			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CCAR conditional Non-objection (post)	0.123* (1.75)		0.092 (1.45)		0.153* (1.84)		0.127 (1.58)	
CCAR objection (post)	0.191*** (3.79)		0.112 (1.21)		0.235*** (3.54)		0.289*** (4.31)	
Any objection (post)		0.147** (2.44)		0.098 (1.64)		0.183** (2.45)		0.163** (2.07)
CCAR	-0.097** (-2.11)	-0.096** (-2.10)	0.026 (0.57)	0.028 (0.62)				
LISCC	0.235*** (5.28)	0.233*** (5.23)	-0.011 (-0.21)	-0.010 (-0.20)	0.211*** (3.18)	0.206*** (3.15)	0.000 (0.02)	-0.002 (-0.09)
Lead agent effects	No	No	Yes	Yes	No	No	Yes	Yes
Launch quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1190	1190	1190	1190	607	607	607	607
R-squared	0.3967	0.3956	0.6036	0.6036	0.4741	0.4711	0.5959	0.5931

Table 4: Do banks with greater capitalization vulnerability issue more cov-lites?

This table regresses the fraction of loan amounts each quarter that is cov-lite for each lending institution on capitalization vulnerability evidenced in the DFAST stress tests under the supervisory severely adverse scenario. Capitalization vulnerability is calculated as the DFAST exercise quarter actual capital ratio minus the DFAST projection period minimum capital ratio. The five capital ratios used to calculate capitalization vulnerability are: The common equity tier 1 capital ratio, the tier 1 capital ratio, the total capital ratio, the tier 1 leverage ratio, and the supplementary leverage ratio. The sample period includes the fourth quarter of 2012 through 2021 (the first DFAST exercise was run in the third quarter of 2012). We update banks' DFAST results one quarter after the DFAST test quarter. For banks subject to a biannual testing cycle, we retain DFAST results for up to eight quarters. *LISCC* is an indicator variable that equals one for all LISCC institutions. Only one institution changes LISCC status during the sample period without leaving the sample: UBS AG. Standard errors are adjusted for clustering by lead agent and launch quarter. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Vulnerability CET1	0.065*** (4.44)	0.015 (1.40)								
Vulnerability Tier 1 Capital			0.053*** (4.50)	0.025* (1.77)						
Vulnerability Total Capital					0.049*** (4.41)	0.024* (1.88)				
Vulnerability Tier 1 leverage							0.066*** (3.28)	0.026 (1.24)		
Vulnerability Suppl. leverage									0.133*** (4.09)	0.026 (0.43)
LISCC	0.212*** (3.10)	-0.042 (-1.49)	0.231*** (3.48)	-0.054* (-1.79)	0.240*** (3.58)	-0.051* (-1.92)	0.336*** (4.87)	-0.013 (-0.40)	0.188* (2.06)	-0.067* (-1.89)
Lead agent effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Launch quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	422	422	527	527	527	527	527	527	142	142
R-squared	0.3741	0.5320	0.4150	0.5454	0.4091	0.5458	0.3746	0.5427	0.3542	0.5740

Table 5: Univariate differences in loan liquidity

This table shows univariate differences in loan liquidity for covenant-lite and covenant-heavy loans, respectively, as well as the results from *t*-tests and Wilcoxon rank-sum tests. The sample includes loans issued by non-financial, non-utility US issuers on the US leveraged loan market from 2010 through 2021. Loan data are from Capital IQ LCD. A loan is defined as covenant-lite if it contains a covenant-lite facility, even if other facilities are present. All financial ratios have been winsorized at the 1% and 99% levels. ***, **, and * indicate statistical significance at 1%, 5%, and 10%. Panel A compares covenant-lite loans with covenant-heavy loans, while Panel B compares covenant-lite loans of unregistered firms with covenant-lite loans of registered firms.

	Obs.	Covenant-lite Mean (St.dev.)	Median	Obs.	Covenant-heavy Mean (St.dev.)	Median	Difference in means (<i>t</i> -statistic)	Difference in medians (<i>z</i> -statistic)
Traded on secondary market	3434	0.832 (0.374)	1.000	3596	0.467 (0.499)	0.000	0.365*** (34.797)	1.000*** (31.963)
Time-to-break	2844	12.162 (5.702)	12.000	1676	16.626 (9.246)	15.000	-4.464*** (-17.865)	-3.000*** (-18.101)
Initial # LP Funds	3430	4.974 (6.370)	2.000	3403	1.781 (3.601)	0.000	3.192*** (25.528)	2.000*** (29.194)
Average % principal traded	2839	1.251 (1.927)	0.655	1570	0.618 (1.016)	0.338	0.633*** (14.282)	0.317*** (15.727)
% months with trade	2839	51.048 (30.920)	47.619	1570	32.048 (23.928)	25.536	19.000*** (22.687)	22.083*** (19.793)
Bid-ask spread (3 month average)	1824	0.653 (0.434)	0.537	1183	0.774 (0.382)	0.701	-0.121*** (-8.048)	-0.165*** (-15.015)
Bid-ask spread (1 st year average)	1893	0.779 (0.529)	0.641	1295	0.920 (0.571)	0.781	-0.141*** (-7.034)	-0.140*** (-12.267)
Bid-ask spread (2 nd year average)	1779	1.128 (1.117)	0.800	1260	1.123 (1.112)	0.851	0.005 (0.129)	-0.051* (-1.910)
Bid-ask spread (3 rd year average)	1320	1.386 (1.934)	0.891	959	1.421 (1.895)	0.892	-0.034 (-0.421)	-0.001 (0.506)
Number of quotes (3 month average)	1824	2.911 (1.913)	2.275	1183	2.090 (1.518)	1.593	0.821*** (13.064)	0.682*** (14.308)

Number of quotes (1 st year average)	1893	3.189 (2.035)	2.590	1295	2.242 (1.574)	1.734	0.947*** (14.792)	0.855*** (15.893)
Number of quotes (2 nd year average)	1779	3.264 (2.007)	2.724	1260	2.185 (1.528)	1.686	1.079*** (16.814)	1.039*** (17.881)
Number of quotes (3 rd year average)	1320	3.260 (2.032)	2.695	959	2.134 (1.490)	1.673	1.126*** (15.266)	1.022*** (15.950)
Total observations	7030							

Table 6: Are covenant-lite loans more likely to trade?

This table reports average marginal effects from probit regressions predicting whether a loan trades on the secondary market. The dependent variable equals one if LCD records a break price for the loan, indicating that the loan broke into secondary market trading. The sample period is 2010 through 2021. Only those loans that have an institutional tranche are included in the regressions. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	All firms		Rated firms	Registered firms	
	(1)	(2)	(3)	(4)	(5)
Covenant lite loan	0.045*** (3.05)	0.044*** (2.62)	0.040*** (2.75)	0.047*** (2.95)	0.045*** (2.76)
Covenant lite x Registered		0.001 (0.08)			
Registered		0.008 (0.58)			
Ln(Assets)					0.002 (0.28)
Leverage					0.023 (0.82)
Current ratio					0.017*** (2.65)
Ln(1+Coverage Ratio)					-0.012* (-1.74)
Tangibility					-0.011 (-0.31)
Ln(Maturity)	0.088*** (3.42)	0.089*** (3.43)	0.102*** (4.02)	0.106*** (3.34)	0.112*** (3.69)
Ln(Deal size)	0.141*** (17.93)	0.140*** (18.03)	0.125*** (16.36)	0.089*** (8.66)	0.087*** (8.01)
Sponsored	0.023* (1.69)	0.026* (1.76)	0.019 (1.29)	0.006 (0.39)	0.004 (0.23)
LBO	0.007 (0.47)	0.007 (0.47)	0.019 (1.15)	-0.019 (-0.69)	-0.006 (-0.22)
Revolver	-0.015 (-0.89)	-0.015 (-0.84)	-0.001 (-0.09)	-0.020 (-1.20)	-0.018 (-1.02)

Term loan A	-0.048 (-1.64)	-0.047 (-1.63)	-0.016 (-0.46)	0.000 (0.00)	0.010 (0.34)
Asset based loan	0.021 (0.73)	0.020 (0.73)	0.019 (0.66)	0.013 (0.38)	0.012 (0.37)
Second lien facility	-0.009 (-0.63)	-0.009 (-0.58)	-0.006 (-0.38)	0.037 (1.30)	0.045 (1.55)
Ln(Firm age)	0.002 (0.37)	0.002 (0.29)	0.001 (0.29)	-0.003 (-0.41)	-0.002 (-0.29)
Rating effects	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Observations	5646	5646	5152	1982	1982

Table 7: Do covenant-lite loans break into secondary market trading more quickly?

The dependent variable is time-to-break, the number of days between the launch date of the loan and the first day on which the loan is traded in the loan trading market. Non-traded loans are excluded from the regression. The average is 14 days. The sample period is 2010 through 2021. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	All firms		Rated firms	Registered firms	
	(1)	(2)	(3)	(4)	(5)
Covenant lite loan	-2.840*** (-8.23)	-3.586*** (-8.16)	-2.666*** (-7.29)	-1.937*** (-3.43)	-2.151*** (-3.91)
Covenant lite x Registered		1.658*** (3.24)			
Registered		-1.615*** (-2.96)			
Ln(Assets)					-0.970*** (-4.54)
Leverage					-4.170*** (-5.99)
Current ratio					0.001 (0.00)
Ln(1+Coverage Ratio)					0.015 (0.07)
Tangibility					0.355 (0.25)
Ln(Maturity)	1.876** (2.61)	1.831** (2.58)	1.069 (1.46)	3.117*** (3.02)	1.745* (1.76)
Ln(Deal size)	-0.522*** (-3.10)	-0.439*** (-2.75)	-0.518*** (-3.14)	-0.553** (-2.40)	0.252 (1.03)
Sponsored	-0.975*** (-3.04)	-1.174*** (-3.66)	-0.998*** (-3.04)	-1.371*** (-3.08)	-0.813* (-1.99)
LBO	0.591** (2.03)	0.620** (2.13)	0.616* (1.98)	0.942 (1.56)	0.267 (0.42)
Revolver	4.154*** (13.74)	4.026*** (13.66)	4.335*** (14.97)	3.526*** (8.41)	2.606*** (6.28)

Term loan A	1.887** (2.67)	1.952*** (2.74)	1.959*** (2.84)	3.246*** (4.16)	2.703*** (3.49)
Asset based loan	0.394 (1.03)	0.434 (1.12)	0.399 (1.00)	1.014 (1.20)	0.991 (1.18)
Second lien facility	1.460*** (4.18)	1.439*** (4.13)	1.475*** (4.15)	2.131*** (2.73)	2.293*** (2.86)
Ln(Firm age)	-0.344** (-2.54)	-0.311** (-2.20)	-0.375*** (-2.81)	-0.442* (-1.76)	-0.229 (-0.94)
Rating effects	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Observations	4463	4463	4282	1743	1743
R-squared	0.2767	0.2802	0.2610	0.2950	0.3259

Table 8: Do covenant-lite loans trade more frequently?

The dependent variables are the average percentage of a loan's principal that is traded monthly and the percentage of months in which at least one loan participation fund trades the loan. Only trades by loan participation funds are tracked. The sample period is June 30, 2010 through December 31, 2021. Loans were matched to the CRSP mutual fund database based on the security name indicated in CRSP. *Initial # LP Funds* indicates the number of loan participation funds that hold shares in the loan after its first month of trading. Loans that are never held by any loan participation fund are excluded. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

Panel A: All trades

Dep. variable:	Monthly average % principal traded					% months with trade				
	All firms (1)	(2)	Rated firms (3)	Registered firms (4)	(5)	All firms (6)	(7)	Rated firms (8)	Registered firms (9)	(10)
Covenant lite loan	0.275*** (4.59)	0.461*** (5.23)	0.261*** (4.37)	0.156** (2.25)	0.181** (2.63)	3.179*** (3.21)	4.136*** (3.03)	2.740*** (2.71)	2.860** (2.30)	3.085** (2.43)
Covenant lite x Registered		-0.377*** (-4.17)					-1.917 (-1.16)			
Registered		0.208*** (3.14)					-0.040 (-0.03)			
# LP Funds	0.078*** (6.09)	0.079*** (6.19)	0.078*** (6.01)	0.048*** (4.41)	0.049*** (4.42)	1.318*** (11.84)	1.328*** (11.92)	1.291*** (11.72)	0.957*** (8.75)	0.956*** (8.76)
Ln(Assets)					0.161*** (3.61)					-0.028 (-0.06)
Leverage					0.307** (2.28)					1.461 (0.62)
Current ratio					-0.026 (-0.90)					0.376 (0.75)
Ln(1+Coverage Ratio)					-0.007 (-0.24)					-0.735 (-1.20)

Tangibility					-0.112 (-0.66)					5.108 (1.42)
Ln(Maturity)	-0.530*** (-3.14)	-0.525*** (-3.10)	-0.505** (-2.53)	-0.671** (-2.35)	-0.442 (-1.56)	4.683* (1.76)	4.612* (1.74)	7.189** (2.53)	5.633 (1.43)	5.956 (1.52)
Ln(Deal size)	-0.424*** (-6.34)	-0.427*** (-6.38)	-0.425*** (-6.29)	-0.284*** (-4.01)	-0.399*** (-4.67)	6.239*** (11.06)	6.300*** (11.09)	6.308*** (11.37)	5.073*** (6.43)	5.032*** (6.59)
Sponsored	0.057 (0.87)	0.054 (0.82)	0.069 (1.01)	0.035 (0.40)	-0.030 (-0.32)	-1.985** (-2.17)	-2.309** (-2.44)	-1.710* (-1.85)	-3.764*** (-2.91)	-3.933*** (-2.96)
LBO	-0.112* (-1.94)	-0.120** (-2.04)	-0.123** (-2.04)	-0.004 (-0.05)	0.057 (0.58)	4.538*** (3.60)	4.492*** (3.59)	4.271*** (3.29)	9.067*** (3.76)	9.854*** (3.71)
Revolver	-0.349*** (-5.35)	-0.338*** (-5.21)	-0.361*** (-5.37)	-0.018 (-0.21)	0.118 (1.36)	-1.460 (-1.36)	-1.511 (-1.42)	-1.406 (-1.26)	0.462 (0.35)	0.563 (0.43)
Term loan A	0.127 (1.37)	0.086 (0.96)	0.117 (1.23)	-0.216** (-2.59)	-0.176** (-2.14)	-8.889*** (-5.41)	-9.083*** (-5.61)	-8.474*** (-4.87)	-10.955*** (-5.24)	-10.652*** (-5.22)
Asset based loan	0.171*** (3.21)	0.157*** (3.05)	0.146** (2.61)	-0.034 (-0.42)	-0.046 (-0.56)	-0.489 (-0.42)	-0.549 (-0.47)	-0.607 (-0.53)	-4.248* (-1.72)	-4.179 (-1.67)
Second lien facility	0.227** (2.66)	0.214** (2.47)	0.234** (2.67)	0.012 (0.10)	-0.021 (-0.17)	-1.646 (-1.63)	-1.880* (-1.81)	-1.799* (-1.77)	-5.142** (-2.08)	-4.702* (-1.97)
Ln(Firm age)	-0.029 (-1.13)	-0.029 (-1.14)	-0.031 (-1.17)	0.020 (0.64)	-0.017 (-0.53)	-0.255 (-0.65)	-0.198 (-0.49)	-0.196 (-0.49)	0.101 (0.18)	0.109 (0.19)
Rating effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4351	4351	4202	1764	1764	4351	4351	4202	1764	1764
R-squared	0.2995	0.3022	0.2994	0.3554	0.3719	0.5344	0.5349	0.5336	0.5375	0.5393

Panel B: Purchases only

Dep. variable:	Monthly average % principal bought					% months with purchase				
	All firms (1)	All firms (2)	Rated firms (3)	Registered firms (4)	Registered firms (5)	All firms (6)	All firms (7)	Rated firms (8)	Registered firms (9)	Registered firms (10)
Covenant lite loan	0.138*** (4.83)	0.228*** (5.28)	0.131*** (4.57)	0.073** (2.42)	0.086*** (2.85)	1.753*** (2.71)	3.048*** (3.13)	1.493** (2.28)	1.025 (1.27)	1.269 (1.56)
Covenant lite x Registered		-0.183*** (-4.13)					-2.618** (-2.25)			
Registered		0.094*** (2.96)					1.108 (1.20)			
# LP Funds	0.026*** (6.70)	0.027*** (6.84)	0.026*** (6.64)	0.015*** (3.84)	0.015*** (3.85)	1.374*** (14.13)	1.382*** (14.42)	1.350*** (13.92)	0.897*** (8.25)	0.899*** (8.21)
Ln(Assets)					0.078*** (3.49)					0.614 (1.62)
Leverage					0.136** (2.30)					1.705 (0.92)
Current ratio					-0.017 (-1.18)					0.157 (0.43)
Ln(1+Coverage Ratio)					-0.004 (-0.28)					-0.597 (-1.35)
Tangibility					-0.044 (-0.57)					3.297 (1.16)
Ln(Maturity)	-0.186** (-2.16)	-0.184** (-2.14)	-0.172* (-1.68)	-0.268 (-1.61)	-0.158 (-0.92)	0.509 (0.28)	0.514 (0.28)	1.945 (1.02)	2.855 (0.99)	3.949 (1.33)
Ln(Deal size)	-0.160*** (-6.75)	-0.161*** (-6.78)	-0.159*** (-6.69)	-0.099*** (-3.66)	-0.155*** (-4.40)	5.904*** (14.68)	5.904*** (14.91)	6.080*** (14.99)	5.952*** (9.50)	5.481*** (8.71)

Sponsored	0.058* (1.83)	0.055* (1.77)	0.064* (1.93)	0.025 (0.66)	-0.006 (-0.15)	0.698 (0.99)	0.580 (0.79)	1.035 (1.44)	-0.626 (-0.62)	-0.964 (-0.94)
LBO	-0.060** (-2.28)	-0.064** (-2.40)	-0.066** (-2.39)	0.022 (0.53)	0.050 (1.06)	2.101** (2.43)	2.045** (2.37)	1.907** (2.10)	7.689*** (3.89)	8.425*** (4.10)
Revolver	-0.187*** (-5.53)	-0.182*** (-5.37)	-0.193*** (-5.51)	-0.036 (-1.03)	0.030 (0.81)	-3.229*** (-4.16)	-3.182*** (-4.19)	-3.379*** (-4.32)	-0.946 (-0.93)	-0.352 (-0.35)
Term loan A	0.052 (1.21)	0.032 (0.78)	0.047 (1.07)	-0.121*** (-2.80)	-0.102** (-2.40)	-6.664*** (-4.95)	-6.944*** (-5.13)	-6.528*** (-4.72)	-9.168*** (-5.11)	-8.812*** (-4.94)
Asset based loan	0.095*** (3.30)	0.088*** (3.18)	0.086*** (2.82)	-0.002 (-0.05)	-0.008 (-0.20)	-0.850 (-0.92)	-0.946 (-1.03)	-0.826 (-0.87)	-3.803* (-1.94)	-3.816* (-1.91)
Second lien facility	0.089** (2.35)	0.081** (2.13)	0.092** (2.35)	-0.019 (-0.36)	-0.036 (-0.64)	-1.333* (-1.75)	-1.476* (-1.91)	-1.421* (-1.85)	-5.212*** (-2.73)	-5.033*** (-2.71)
Ln(Firm age)	-0.014 (-1.18)	-0.014 (-1.16)	-0.015 (-1.23)	0.015 (0.99)	-0.003 (-0.24)	-0.188 (-0.60)	-0.170 (-0.52)	-0.203 (-0.62)	0.087 (0.20)	-0.065 (-0.15)
Rating effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4351	4351	4202	1764	1764	4351	4351	4202	1764	1764
R-squared	0.2168	0.2201	0.2159	0.2475	0.2707	0.5555	0.5562	0.5537	0.5592	0.5616

Table 9: Do secondary market quotes indicate more liquidity for covenant-lite loans?

In Panel A, the dependent variable is the average of the difference between the average bid quote and the average ask quote for a loan divided by the average ask quote multiplied by 100. In Panel B, the dependent variable is the average number of brokers that supply a quote to the Thomson Reuters/LSTA Mark-to-Market pricing service. Four time periods are examined: the first three months after the loan's launch date, the first year, the second year, and the third year. The sample includes only those loans that were originated between January 2010 and December 2021 and are tracked by the Thomson Reuters/LSTA Mark-to-Market pricing service. If more than one of a loan's tranches is traded, we use the bid-ask spread of the largest tranche. *Initial # LP Funds* indicates the number of loan participation funds that hold shares in the loan after its first month of trading. If a loan is tracked by LSTA, but no loan participation fund reports holding the loan in CRSP, *Initial # LP Funds* is set to zero for that loan. *Revolver*, *Term Loan A*, *Term Loan C*, *Term Loan D or other*, and *Second Lien* are indicator variables capturing the type of the loan tranche that is tracked in LSTA. Term Loans B act as the base group. Columns (1) through (4) include all loans, columns (5) through (8) exclude loans whose bid-ask midpoint is below 90 cents on the dollar at some point during the respective period as well as debtor-in-possession loans. All bid-ask spreads are winsorized at the 1st and 99th percentiles. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

Panel A: Bid-ask spreads

	All loans				Non-distressed loans			
	3 months	Year 1	Year 2	Year 3	3 months	Year 1	Year 2	Year 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Covenant lite loan	-0.095*** (-5.74)	-0.126*** (-4.62)	-0.064 (-1.04)	-0.097 (-0.67)	-0.092*** (-5.53)	-0.102*** (-4.82)	-0.122*** (-3.59)	-0.114** (-2.25)
Covenant lite x Registered	0.077*** (3.44)	0.099*** (2.99)	-0.007 (-0.11)	-0.138 (-0.71)	0.065*** (3.15)	0.073*** (3.09)	0.054 (1.40)	0.046 (0.96)
Registered	-0.049** (-2.58)	-0.081*** (-2.76)	-0.066 (-1.32)	0.005 (0.03)	-0.045** (-2.43)	-0.071*** (-3.61)	-0.062** (-2.11)	-0.061* (-1.71)
Initial # LP Funds	-0.007*** (-4.98)	-0.008*** (-5.60)	-0.011*** (-4.22)	-0.011** (-2.03)	-0.006*** (-5.75)	-0.006*** (-5.39)	-0.007*** (-4.05)	-0.006** (-2.38)
Ln(Maturity)	-0.002 (-0.06)	0.057 (1.27)	0.054 (0.64)	-0.063 (-0.26)	-0.048* (-1.71)	-0.042 (-1.23)	-0.038 (-0.77)	-0.063 (-0.49)

Ln(Deal size)	-0.072*** (-7.43)	-0.083*** (-6.51)	-0.079*** (-3.25)	-0.138*** (-3.11)	-0.079*** (-9.00)	-0.087*** (-6.81)	-0.083*** (-5.29)	-0.082*** (-3.41)
Sponsored	-0.013 (-0.92)	-0.026 (-1.38)	-0.047 (-1.00)	-0.065 (-0.71)	-0.007 (-0.57)	-0.030* (-2.00)	-0.052** (-2.34)	-0.056* (-1.99)
LBO	0.009 (0.65)	0.032* (1.76)	0.041 (1.08)	0.041 (0.47)	0.004 (0.29)	0.023 (1.34)	0.051** (2.44)	0.014 (0.42)
Revolver	0.748*** (6.24)	0.965*** (5.46)	1.250*** (4.55)	0.843** (2.13)	0.690*** (5.96)	0.881*** (6.47)	1.251*** (4.72)	1.328*** (2.77)
Term Loan A	0.124*** (2.87)	0.181*** (2.75)	0.246** (2.59)	-0.040 (-0.35)	0.130*** (3.01)	0.163*** (2.91)	0.210*** (3.08)	0.141* (1.95)
Asset based loan	0.013 (0.79)	0.049* (1.74)	0.109** (2.13)	0.030 (0.34)	0.033** (2.26)	0.022 (1.08)	0.013 (0.41)	0.050 (1.27)
Second lien	0.177*** (3.62)	0.194*** (2.89)	0.408*** (2.98)	0.924** (2.53)	0.184*** (3.77)	0.239*** (4.15)	0.285*** (3.74)	0.023 (0.28)
Ln(Firm age)	-0.016*** (-2.93)	-0.033*** (-4.12)	-0.063*** (-3.75)	-0.093** (-2.62)	-0.016*** (-3.20)	-0.023*** (-3.56)	-0.023** (-2.08)	-0.038** (-2.06)
Rating effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2895	3067	2922	2183	2838	2754	2306	1586
R-squared	0.4229	0.4322	0.3076	0.2272	0.3998	0.4115	0.3006	0.2617

Panel B: Number of quotes

	All loans				Non-distressed loans			
	3 months	Year 1	Year 2	Year 3	3 months	Year 1	Year 2	Year 3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Covenant lite loan	0.447*** (5.68)	0.467*** (6.12)	0.406*** (5.14)	0.171* (1.99)	0.459*** (5.93)	0.487*** (5.64)	0.446*** (4.34)	0.224* (1.87)
Covenant lite x Registered	-0.330*** (-3.03)	-0.286** (-2.47)	-0.065 (-0.48)	-0.010 (-0.07)	-0.357*** (-3.35)	-0.306** (-2.49)	-0.061 (-0.39)	-0.011 (-0.06)
Registered	0.321*** (3.20)	0.316*** (3.79)	0.202 (1.67)	0.197 (1.60)	0.328*** (3.33)	0.341*** (3.65)	0.175 (1.33)	0.230 (1.46)
Initial # LP Funds	0.068*** (7.76)	0.072*** (7.93)	0.060*** (8.50)	0.054*** (5.64)	0.069*** (8.07)	0.080*** (8.10)	0.063*** (7.55)	0.056*** (4.46)
Ln(Maturity)	-0.548*** (-2.77)	-0.564*** (-2.72)	-0.427** (-2.40)	-0.145 (-0.49)	-0.647*** (-2.95)	-0.714*** (-2.93)	-0.575** (-2.46)	0.025 (0.08)
Ln(Deal size)	0.934*** (12.21)	1.030*** (13.65)	1.031*** (15.35)	1.023*** (14.66)	0.951*** (12.25)	1.030*** (12.78)	1.095*** (14.77)	1.005*** (13.08)
Sponsored	0.146* (1.87)	0.185** (2.44)	0.292*** (3.81)	0.324*** (3.40)	0.132* (1.68)	0.186** (2.19)	0.298*** (2.99)	0.356*** (2.96)
LBO	-0.194*** (-2.91)	-0.283*** (-3.93)	-0.312*** (-4.64)	-0.277*** (-3.08)	-0.184*** (-2.73)	-0.263*** (-3.36)	-0.303*** (-3.89)	-0.204 (-1.65)
Revolver	-1.956*** (-5.15)	-2.555*** (-6.89)	-2.790*** (-8.25)	-2.847*** (-8.28)	-1.993*** (-7.50)	-2.550*** (-8.07)	-2.827*** (-8.61)	-2.898*** (-8.17)
Term Loan A	-1.541*** (-7.70)	-1.647*** (-10.00)	-1.581*** (-9.20)	-1.644*** (-7.22)	-1.578*** (-7.77)	-1.767*** (-9.47)	-1.627*** (-8.65)	-1.603*** (-6.48)

Asset based loan	-0.324*** (-3.42)	-0.403*** (-4.45)	-0.454*** (-5.51)	-0.500*** (-4.06)	-0.349*** (-3.65)	-0.396*** (-3.85)	-0.477*** (-4.38)	-0.636*** (-3.95)
Second lien	-0.493*** (-2.77)	-0.519*** (-3.36)	-0.477*** (-2.90)	-0.327** (-2.12)	-0.479** (-2.64)	-0.535*** (-3.24)	-0.561*** (-3.12)	-0.331* (-1.77)
Ln(Firm age)	0.107*** (3.21)	0.106*** (3.29)	0.081*** (2.83)	0.069* (1.85)	0.109*** (3.23)	0.115*** (3.50)	0.099** (2.69)	0.094* (1.99)
Rating effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2895	3067	2922	2183	2838	2754	2306	1586
R-squared	0.4531	0.5054	0.5037	0.4931	0.4605	0.5190	0.5249	0.4958

Table 10: Differences in spreads cov-lite vs. cov-heavy

This table presents regressions of the credit spread paid on a loan's first lien institutional tranche on an indicator of whether or not the institutional loan is cov-lite. Only loans that have both an institutional tranche and a revolving line of credit are included. The sample period is 2010 through 2021. The sample is restricted to loans rated between BB+ and B- since 93% of cov-lite loans have a rating within this range. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	All sample firms			Registered firms	
	(1)	(2)	(3)	(4)	(5)
Covenant lite loan	-36.031*** (-5.07)	-15.488** (-2.52)	-22.931*** (-3.57)	-8.495 (-0.76)	-9.661 (-0.84)
Cov-lite x registered			17.601** (2.33)		
Registered			-11.144 (-1.60)		
Ln(Assets)					-1.734 (-0.60)
Leverage					-20.013 (-1.35)
Current ratio					2.563 (0.75)
Ln(1+Coverage Ratio)					-2.335 (-0.52)
Tangibility					28.806 (1.42)
Revolver spread	0.502*** (13.53)	0.684*** (18.88)	0.680*** (18.61)	0.638*** (10.65)	0.635*** (10.70)
Ln (Revolver Maturity)		29.678 (1.32)	28.805 (1.27)	27.308 (0.85)	27.430 (0.81)
Ln (Institutional Maturity)		-67.713** (-2.37)	-68.150** (-2.41)	-74.002* (-1.96)	-79.054** (-2.05)
Ln (Revolver Amount)		5.451 (1.41)	5.621 (1.47)	2.066 (0.47)	2.590 (0.55)

Ln (Institutional Amount)		-17.062*** (-3.81)	-17.076*** (-3.84)	-10.582* (-1.79)	-8.673 (-1.40)
Sponsored		-0.590 (-0.11)	-1.829 (-0.34)	-8.985 (-0.97)	-6.807 (-0.81)
LBO		-4.078 (-1.32)	-3.484 (-1.13)	3.531 (0.44)	0.915 (0.12)
Term loan A		14.621 (1.51)	14.968 (1.57)	0.450 (0.05)	1.207 (0.13)
Asset based loan		154.760*** (12.46)	153.881*** (12.50)	125.146*** (5.61)	124.169*** (5.48)
Second lien facility		-14.472*** (-4.65)	-13.814*** (-4.55)	-6.086 (-0.74)	-4.478 (-0.56)
Ln(Firm age)		-0.286 (-0.15)	-0.314 (-0.17)	1.724 (0.44)	1.851 (0.45)
Rating effects	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes
Observations	1754	1727	1727	624	624
R-squared	0.5973	0.7227	0.7240	0.7247	0.7280

Table 11: Institutional spread, revolver spread, bid-ask spread, and cov-lite

This table presents regressions examining the relation between the credit spread paid on a loan's first lien institutional tranche, the credit spread paid on the loan's revolving line of credit, quoted bid-ask spread, and an indicator of whether or not the institutional loan is cov-lite. The dependent variable is the institutional credit spread. The sample is restricted to loans that have both a first lien institutional tranche with quoted bid-ask spread data as well as a revolving line of credit. The sample period is 2010 through 2021. The sample is restricted to loans rated between BB+ and B- since 93% of cov-lite loans have a rating within this range. All bid-ask spreads are winsorized at the 1st and 99th percentiles. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Covenant lite loan	-28.232*** (-3.28)	-17.441 (-1.50)	-29.467*** (-3.31)	-17.122 (-1.28)	-29.690*** (-3.41)	-15.756 (-1.36)	-31.291*** (-3.43)	-24.974** (-2.37)	-27.881** (-2.66)	-17.809 (-1.22)
Covenant lite x Registered	12.638 (1.38)		13.689 (1.49)		14.095 (1.57)		11.217 (1.05)		13.382 (1.12)	
Registered	-5.495 (-0.65)		-8.570 (-0.95)		-7.553 (-0.92)		-3.584 (-0.37)		-6.789 (-0.55)	
Revolver spread	0.644*** (15.73)	0.622*** (7.70)	0.591*** (13.07)	0.585*** (7.25)	0.583*** (14.37)	0.555*** (8.27)	0.605*** (14.97)	0.560*** (7.35)	0.644*** (13.65)	0.604*** (6.93)
# LP Funds	-1.731*** (-4.45)	-1.165* (-1.85)	-1.584*** (-3.49)	-0.633 (-0.87)	-1.531*** (-3.54)	-0.553 (-0.85)	-1.641*** (-4.06)	-1.077 (-1.64)	-1.533*** (-3.14)	-0.548 (-0.57)
3-month bid-ask Spread			58.815*** (4.06)	59.191** (2.64)						
1-year bid-ask Spread					47.703*** (5.06)	70.275*** (4.89)				
2-year bid-ask Spread							15.022*** (5.22)	23.372*** (3.34)		

3-year bid-ask Spread									8.335*** (3.58)	15.437** (2.27)
Ln(Assets)		-0.148 (-0.04)		-0.014 (-0.00)		-0.590 (-0.17)		-1.874 (-0.50)		0.004 (0.00)
Leverage		-46.785** (-2.13)		-40.694* (-1.71)		-40.771* (-1.87)		-46.779* (-1.97)		-57.269 (-1.67)
Current ratio		4.158 (1.05)		3.782 (1.03)		3.852 (1.06)		2.781 (0.68)		3.553 (0.57)
Ln(1+Coverage) Ratio		-4.763 (-1.03)		-6.707 (-1.23)		-4.118 (-0.82)		-4.538 (-0.78)		-10.850 (-1.38)
Tangibility		37.801 (1.33)		49.014 (1.59)		51.565 (1.66)		40.887 (1.36)		56.896 (1.27)
Ln(Revolver term)	17.960 (0.65)	52.610 (1.36)	29.656 (1.06)	62.494 (1.55)	23.261 (0.87)	47.668 (1.34)	-0.532 (-0.02)	17.245 (0.41)	-1.546 (-0.05)	-12.070 (-0.34)
Ln(Institutional term)	-90.763*** (-3.52)	-153.914*** (-2.99)	-88.996*** (-3.60)	-153.737*** (-3.02)	-90.490*** (-3.64)	-155.873*** (-3.26)	-74.015*** (-2.88)	-134.675** (-2.53)	-111.723*** (-2.78)	-208.904** (-2.61)
Ln(Revolver amount)	6.415 (1.31)	5.106 (0.90)	5.856 (1.16)	4.939 (0.77)	6.676 (1.31)	5.650 (1.01)	4.869 (0.98)	3.794 (0.65)	7.700 (1.37)	3.963 (0.43)
Ln(Institutional amount)	-11.627* (-1.88)	-7.800 (-0.84)	-6.319 (-0.98)	-4.390 (-0.44)	-8.360 (-1.35)	-4.182 (-0.43)	-11.036* (-1.72)	-6.577 (-0.64)	-10.240 (-1.28)	-10.733 (-1.03)
Sponsored	1.220 (0.18)	0.414 (0.04)	-0.225 (-0.03)	1.251 (0.11)	1.372 (0.20)	0.487 (0.04)	2.075 (0.30)	1.449 (0.12)	2.780 (0.31)	9.473 (0.60)
LBO	-7.267*** (-2.71)	-10.175 (-1.08)	-3.667 (-1.24)	-12.227 (-1.17)	-5.643** (-2.07)	-10.101 (-1.03)	-6.084* (-1.93)	-6.790 (-0.57)	-7.924** (-2.44)	-5.094 (-0.33)
Term loan A	-6.844 (-0.60)	-0.164 (-0.01)	-12.172 (-0.99)	-0.118 (-0.01)	-12.781 (-1.12)	-1.272 (-0.09)	-10.801 (-1.04)	-5.135 (-0.40)	-6.979 (-0.56)	6.818 (0.39)

Asset loan	based	151.772*** (11.35)	130.870*** (4.68)	140.349*** (9.80)	122.265*** (4.10)	137.550*** (10.34)	119.545*** (4.61)	141.401*** (10.72)	117.661*** (4.26)	151.781*** (9.48)	133.461*** (4.30)
Second facility	lien	-7.857** (-2.29)	-0.397 (-0.04)	-8.878** (-2.22)	-2.665 (-0.21)	-8.130** (-2.15)	-0.793 (-0.07)	-8.715** (-2.58)	-0.884 (-0.07)	-3.122 (-0.84)	2.449 (0.14)
Ln(Firm age)		-1.201 (-0.47)	-2.125 (-0.43)	0.405 (0.14)	-3.670 (-0.60)	-0.575 (-0.20)	-4.374 (-0.85)	-1.345 (-0.50)	-4.227 (-0.84)	-5.247* (-2.02)	-3.030 (-0.67)
Rating effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations		1145	405	1016	357	1063	373	1054	363	774	259
R-squared		0.7440	0.7728	0.7506	0.7860	0.7527	0.7978	0.7484	0.7890	0.7798	0.8192

Table 12: Instrumenting bid-ask spreads

This table presents both the first and second stage of instrumental variables regressions examining the relation between the credit spread paid on a loan's first lien institutional tranche and the quoted bid-ask spread. The quoted bid-ask spread for a loan is instrumented using the average quoted bid-ask spread in the borrower's industry immediately prior to loan origination. The sample includes loans originated from 2010 through 2021 that have a first lien institutional tranche with quoted bid-ask spread data. The sample is restricted to loans rated between BB+ and B- since 93% of cov-lite loans have a rating within this range. The industry-level revolver spread represents the industry average interest rate spread over LIBOR during the quarter preceding loan origination. All bid-ask spreads are winsorized at the 1st and 99th percentiles. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	First stage		Second stage	
	3-month bid-ask spread	1-year bid-ask spread	Institutional spread	Institutional spread
	(1)	(2)	(3)	(4)
Industry average bid-ask spread	0.034*** (2.94)	0.049*** (3.68)		
3-month bid-ask spread			492.833** (2.44)	
1-year bid-ask spread				326.605* (1.97)
Covenant lite loan	-0.084*** (-4.64)	-0.116*** (-3.41)	-36.069* (-1.71)	-39.70* (-1.74)
Covenant lite x Registered	0.071*** (2.80)	0.067* (1.70)	2.461 (0.13)	18.259 (1.06)
Registered	-0.022 (-0.99)	-0.028 (-0.82)	-26.855** (-2.43)	-32.266*** (-2.88)
Industry average revolver spread	-0.000 (-0.80)	-0.000 (-0.18)	0.010 (0.28)	-0.017 (-0.45)
# LP Funds	-0.006*** (-4.30)	-0.008*** (-6.06)	-0.429 (-0.32)	-0.573 (-0.39)
Ln(Institutional	-0.024	0.017	-45.249**	-63.852**

term)	(-0.68)	(0.35)	(-2.10)	(-2.64)
Ln(Institutional amount)	-0.080*** (-6.65)	-0.089*** (5.25)	35.507** (2.04)	25.184* (1.59)
Sponsored	-0.011 (-0.62)	-0.020 (-0.75)	-6.564 (-0.67)	-6.374 (-0.61)
LBO	0.003 (0.16)	0.023 (1.33)	4.376 (0.54)	0.066 (0.01)
Term loan A	-0.047* (-1.78)	(-0.009) (-0.16)	-13.652 (-0.74)	-36.801* (-1.99)
Asset based loan	0.028 (1.59)	(0.073)*** (3.30)	35.511*** (3.14)	24.238 (1.60)
Second lien facility	0.003 (0.22)	0.005 (0.27)	-6.074 (-0.99)	-6.129 (-0.89)
Ln(Firm age)	-0.012* (-1.91)	-0.031*** (-3.71)	-3.697 (-0.99)	0.391 (0.06)
Industry effects	Yes	Yes	Yes	Yes
Year-quarter x rating effects	Yes	Yes	Yes	Yes
Observations	2059	2145	2059	2145
Weak identification test				
Kleibergen-Paap rk Wald F			8.650	13.510
Stock-Yogo critical values:				
10% maximal IV size:			16.38	16.38
15% maximal IV size:			8.96	8.96
20% maximal IV size:			6.66	6.66
Weak-IV robust inference:				
Anderson-Rubin χ^2			6.88***	5.92**
(p-value)			(0.0087)	(0.0150)
Stock-Wright LM S χ^2			6.81***	6.10**
(p-value)			(0.0091)	(0.0135)

Internet Appendix

This internet appendix contains the following additional analyses and statistics. Table IA1 analyses whether the impact of the leveraged lending guidance is greater for riskier loans across different types of institutions. Table IA2 provides summary statistics for loan and firm characteristics separately for cov-lite vs. cov-heavy loans. Table IA3 examines whether cov-lite loans are more or less likely to result in bankruptcy than cov-heavy loans. Table IA4 replicates Table 12 with one change: In Table 12, we control for the industry-level revolver spread to capture uncontrolled industry-level variation in credit risk over time. About 43% of the loans in these regressions are accompanied by a revolver. In Table IA4, we use the spread on that revolver as a control, where available, and substitute the industry-level revolver spread only for those loans not accompanied by a revolver.

Table IA1: Differential response to leveraged lending guidance events by borrower rating

This table regresses the fraction of loan amounts each quarter that is cov-lite for each lending institution on time and rating indicators and their interactions. The sample period includes the years 2010 (first year in which LISCCs are defined) through 2020 (last year in which CCAR banks are known). The sample is restricted to loans that are rated either B or BB. *Previous year* refers to the period from Quarter 2 of 2011 through Quarter 1 of 2012. *Post proposal* covers the period after publication of the proposed guidance but before publication of the final guidance (Quarter 2 of 2012 through Quarter 1 of 2013). *Post guidance* refers to the period after publication of the final guidance but before publication of the clarification of the guidance (Quarter 2 of 2013 through Quarter 4 of 2014). *Post clarification* refers to the period after publication of the clarification but before the OCC statement on non-enforcement of the guidance (Quarter 1 of 2015 through Quarter 1 of 2018). *Post OCC statement* refers to the period after the OCC statement (Quarter 2 of 2018 onwards). *B rated* is an indicator variable that equals one for B-rated loans and zero for BB rated loans. Columns (1) and (2) are restricted to observations where the institution is a LISCC, columns (3) and (4) include all institutions other than LISCCs, and columns (5) and (6) only include foreign and domestic non-LISCC commercial banks. Standard errors are adjusted for clustering by lead agent and launch quarter. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	LISCCs		Non-LISCCs		Non-LISCC commercial banks	
	(1)	(2)	(3)	(4)	(5)	(6)
Previous year	0.132	0.137	0.000	-0.009	-0.000	-0.008
	(1.57)	(1.58)	(0.00)	(-0.25)	(-0.00)	(-0.10)

Post proposal	0.280** (2.61)	0.287** (2.76)	0.182*** (2.74)	0.188*** (8.15)	0.222** (2.46)	0.244** (2.62)
Post guidance	0.458*** (5.27)	0.453*** (5.31)	0.363*** (3.78)	0.383*** (3.86)	0.320*** (2.95)	0.327** (2.45)
Post clarification	0.551*** (8.34)	0.539*** (8.21)	0.352*** (3.02)	0.330** (2.65)	0.356*** (2.71)	0.352** (2.40)
Post OCC statement	0.618*** (8.29)	0.606*** (8.59)	0.576*** (7.16)	0.543*** (5.50)	0.626*** (6.50)	0.560*** (4.37)
B rated	0.002 (0.06)	0.002 (0.08)	0.011* (1.72)	-0.049 (-1.54)	-0.000 (-0.00)	-0.056 (-0.96)
Previous year x B rated	-0.036 (-0.43)	-0.033 (-0.38)	0.169 (1.36)	0.154 (1.32)	0.222 (1.48)	0.205 (1.14)
Post proposal x B rated	0.177 (1.48)	0.176 (1.50)	-0.063 (-1.04)	-0.070 (-0.71)	-0.134 (-1.21)	-0.147 (-1.07)
Post guidance x B rated	0.139 (1.65)	0.151* (1.82)	-0.061 (-0.68)	-0.084 (-1.20)	-0.026 (-0.24)	-0.080 (-0.65)
Post clarification x B rated	0.134*** (6.15)	0.130*** (6.77)	0.109 (0.90)	0.165 (1.39)	-0.004 (-0.03)	0.056 (0.46)
Post OCC statement x B rated	0.156*** (3.28)	0.148*** (3.41)	0.189*** (3.11)	0.193** (2.53)	0.101 (1.42)	0.174 (1.60)
Lead agent effects	No	Yes	No	Yes	No	Yes
Observations	833	833	570	570	415	415
R-squared	0.3929	0.4349	0.2789	0.5150	0.2577	0.4819

Table IA2: Summary statistics

This table shows summary statistics for covenant-lite and covenant-heavy loans, respectively, as well as the results from *t*-tests and Wilcoxon rank-sum tests for differences between the two groups. The sample includes loans issued by non-financial, non-utility US issuers on the US leveraged loan market from 2010 through 2021. Loan data are from Capital IQ LCD. A loan is defined as covenant-lite if it contains a covenant-lite facility, even if other facilities are present. All financial ratios have been winsorized at the 1% and 99% levels. ***, **, and * indicate statistical significance at 1%, 5%, and 10%. Panel A compares covenant-lite loans with covenant-heavy loans, while Panel B compares covenant-lite loans of unregistered firms with covenant-lite loans of registered firms.

Panel A: Cov-lite vs. cov-heavy

	Obs.	Covenant-lite Mean (St.dev.)	Median	Obs.	Covenant-heavy Mean (St.dev.)	Median	Difference in means (<i>t</i> -statistic)	Difference in medians (<i>z</i> -statistic)
Deal size (million)	3434	902.250 (1070.588)	575.165	3596	673.673 (882.779)	388.407	228.577*** (9.742)	186.758*** (13.611)
Maturity	3434	6.221 (0.953)	6.610	3595	5.327 (1.099)	5.020	0.894*** (36.488)	1.590*** (37.539)
Secured	2598	0.988 (0.110)	1.000	2471	0.837 (0.369)	1.000	0.150*** (19.440)	0.000*** (19.123)
First lien rating	3370	5.313 (1.513)	6.000	2653	4.454 (1.784)	5.000	0.859*** (19.815)	1.000*** (20.518)
Rated	3434	0.973 (0.163)	1.000	3596	0.728 (0.445)	1.000	0.245*** (30.857)	0.000*** (28.514)
Sponsored	3434	0.744 (0.437)	1.000	3596	0.479 (0.500)	0.000	0.264*** (23.647)	1.000*** (22.695)
LBO	3434	0.225 (0.418)	0.000	3596	0.111 (0.314)	0.000	0.114*** (12.875)	0.000*** (12.806)
Revolver	3434	0.512 (0.500)	1.000	3596	0.688 (0.463)	1.000	-0.176*** (-15.294)	0.000*** (-15.072)
Term loan A	3434	0.025 (0.156)	0.000	3596	0.260 (0.439)	0.000	-0.235*** (-30.202)	0.000*** (-27.970)

Second lien facility	3434	0.174 (0.379)	0.000	3596	0.077 (0.267)	0.000	0.097*** (12.324)	0.000*** (12.287)
Asset based loan	3434	0.089 (0.284)	0.000	3596	0.091 (0.287)	0.000	-0.002 (-0.313)	0.000 (-0.312)
Unregistered firm	3434	0.618 (0.486)	1.000	3596	0.432 (0.495)	0.000	0.186*** (15.896)	1.000*** (15.612)
Firm age	3398	43.518 (41.663)	30.000	3570	42.490 (39.786)	29.000	1.028 (1.052)	1.000 (1.495)
Assets (million)	1291	29831.906 (220691.724)	3058.954	2026	18195.605 (160106.818)	2185.947	11636.301 (1.639)	873.007*** (7.125)
Leverage	1272	0.464 (0.269)	0.466	2005	0.422 (0.263)	0.397	0.041*** (4.330)	0.068*** (5.282)
Current ratio	1355	1.975 (1.411)	1.620	2118	1.817 (1.315)	1.529	0.158*** (3.302)	0.091*** (3.643)
Ln(1+Coverage Ratio)	1263	1.890 (1.063)	1.650	2004	2.010 (1.078)	1.769	-0.120*** (-3.137)	-0.119*** (-4.520)
Tangibility	1282	0.214 (0.216)	0.124	2013	0.272 (0.243)	0.187	-0.058*** (-7.115)	-0.063*** (-6.827)
Total observations	7030							

Panel B: Unregistered cov-lite vs. registered cov-lite

	Unregistered covenant-lite			Registered covenant-lite				
	Obs.	Mean (St.dev.)	Median	Obs.	Mean (St.dev.)	Median	Difference in means (<i>t</i> -statistic)	Difference in medians (<i>z</i> -statistic)
Deal size (million)	2122	698.967 (740.614)	496.474	1312	1231.034 (1392.434)	801.541	-532.067*** (-12.769)	-305.066*** (-14.396)
Maturity	2122	6.211 (0.988)	6.650	1312	6.238 (0.894)	6.556	-0.027 (-0.830)	0.094 (0.973)
Secured	1584	0.991 (0.097)	1.000	1014	0.983 (0.128)	1.000	0.007 (1.548)	0.000 (1.644)
First lien rating	2069	5.889 (0.997)	6.000	1301	4.396 (1.726)	5.000	1.493*** (28.374)	1.000*** (26.827)
Rated	2122	0.967 (0.180)	1.000	1312	0.982 (0.131)	1.000	-0.016*** (-2.990)	0.000*** (-2.779)
Sponsored	2122	0.904 (0.294)	1.000	1312	0.484 (0.500)	0.000	0.420*** (27.639)	1.000*** (27.411)
LBO	2122	0.280 (0.449)	0.000	1312	0.136 (0.343)	0.000	0.144*** (10.621)	0.000*** (9.838)
Revolver	2122	0.537 (0.499)	1.000	1312	0.471 (0.499)	0.000	0.066*** (3.749)	1.000*** (3.743)
Term loan A	2122	0.007 (0.084)	0.000	1312	0.054 (0.226)	0.000	-0.047*** (-7.229)	0.000*** (-8.572)
Second lien facility	2122	0.231 (0.422)	0.000	1312	0.082 (0.274)	0.000	0.150*** (12.619)	0.000*** (11.248)
Asset based loan	2122	0.092 (0.290)	0.000	1312	0.082 (0.275)	0.000	0.010 (1.020)	0.000 (1.007)
Firm age	2086	36.325 (32.156)	27.000	1312	54.956 (51.381)	35.000	-18.631*** (-11.765)	-8.000*** (-10.109)
Total observations	3434							

Table IA3: Are covenant-lite borrowers more likely to go bankrupt?

This table reports results from a linear probability model predicting borrower bankruptcy. The dependent variable is an indicator for whether the borrower goes bankrupt within a certain timeframe from loan origination, with time horizons from one through five years. Bankruptcy dates are obtained from Capital IQ. The sample period is 2010 through 2021, except in column (5) which excludes the year 2021 since year 2026 bankruptcy data are not fully available at the time of writing. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	1-year horizon	2-year horizon	3-year horizon	4-year horizon	5-year horizon
Cov-lite loan	0.002 (1.48)	0.004 (1.40)	0.002 (0.41)	0.004 (0.75)	0.005 (0.70)
Cov-lite x registered	0.000 (0.18)	-0.005 (-1.48)	-0.000 (-0.08)	-0.001 (-0.15)	-0.001 (-0.08)
Registered	0.001 (0.35)	0.009*** (2.94)	0.012*** (3.02)	0.025*** (3.88)	0.030*** (3.98)
Ln(Maturity)	-0.010 (-1.38)	-0.013 (-1.18)	-0.017 (-1.42)	-0.019 (-1.50)	-0.028* (-1.95)
Ln(Deal size)	-0.000 (-0.19)	-0.000 (-0.12)	-0.000 (-0.22)	-0.001 (-0.49)	0.001 (0.19)
Sponsored	-0.004** (-2.33)	-0.007*** (-3.11)	-0.018*** (-4.33)	-0.022*** (-3.55)	-0.025*** (-3.46)
LBO	0.000 (0.36)	-0.002 (-1.29)	-0.004 (-1.43)	-0.007 (-1.38)	-0.002 (-0.42)
Revolver	0.000 (0.24)	0.002 (0.81)	0.001 (0.20)	0.004 (0.91)	0.002 (0.29)
Term loan A	-0.003 (-1.13)	-0.005 (-1.30)	-0.010** (-2.30)	-0.014** (-2.67)	-0.014** (-2.04)

Asset based loan	-0.002 (-1.04)	0.003 (0.79)	0.004 (0.78)	0.009 (1.07)	0.012 (1.05)
Second lien facility	-0.001 (-0.96)	-0.001 (-0.22)	-0.003 (-0.73)	0.003 (0.63)	0.003 (0.53)
Ln(Firm age)	0.002* (1.79)	0.001 (0.76)	0.000 (0.09)	-0.004 (-1.39)	-0.003 (-1.14)
Constant	0.044** (2.49)	0.051** (2.36)	0.059** (2.45)	0.092*** (2.97)	0.162*** (4.96)
Rating effects	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes
Year-quarter effects	Yes	Yes	Yes	Yes	Yes
Observations	6929	6929	6929	6929	6131
R-squared	0.0317	0.0379	0.0427	0.0588	0.0781

Table IA4: Instrumenting bid-ask spreads, controlling for deal-level revolver spread where available

This table presents both the first and second stage of instrumental variables regressions examining the relation between the credit spread paid on a loan's first lien institutional tranche and the quoted bid-ask spread. The quoted bid-ask spread for a loan is instrumented using the average quoted bid-ask spread in the borrower's industry immediately prior to loan origination. The sample includes loans originated from 2010 through 2021 that have a first lien institutional tranche with quoted bid-ask spread data. The sample is restricted to loans rated between BB+ and B- since 93% of cov-lite loans have a rating within this range. When the loan is accompanied by a revolving line of credit, we include the spread on the revolver as a control. Otherwise, we use as the revolver spread the industry average interest rate spread over LIBOR during the quarter preceding loan origination. Standard errors are adjusted for clustering by firm and time. ***, **, and * indicate statistical significance at 1%, 5%, and 10%.

	First stage		Second stage	
	3-month bid-ask spread	1-year bid-ask spread	Institutional spread	Institutional spread
	(1)	(2)	(3)	(4)
Industry average bid-ask spread	0.032** (2.57)	0.048*** (3.60)		
3-month bid-ask spread			430.251** (2.15)	
1-year bid-ask spread				264.854* (1.75)
Covenant lite loan	-0.081*** (-4.61)	-0.107*** (-3.26)	-37.839* (-1.93)	-42.827** (-2.27)
Covenant lite x Registered	0.059** (2.36)	0.056 (1.45)	5.613 (0.34)	16.898 (1.23)
Registered	(-0.007) (-0.36)	(-0.022) (-0.65)	-27.607*** (-2.85)	-26.771*** (2.76)
Revolver spread	0.0002*** (2.88)	0.0003** (2.30)	0.098* (1.70)	0.117* (1.99)
# LP Funds	-0.005*** (-3.54)	-0.008*** (-5.83)	-0.638 (-0.55)	-0.690 (-0.55)

Ln(Institutional term)	-0.026 (-0.79)	0.017 (0.37)	-49.620** (-2.51)	-67.064*** (-3.11)
Ln(Institutional amount)	-0.081*** (-6.56)	-0.087*** (-5.32)	30.120 (1.70)	18.889 (1.31)
Sponsored	-0.014 (-0.88)	-0.019 (-0.73)	-5.823 (-0.67)	-8.364 (-0.91)
LBO	-0.004 (-0.26)	0.019 (0.98)	6.036 (0.85)	0.634 (0.09)
Term loan A	-0.039 (-1.54)	-0.013 (-0.26)	-13.685 (-0.82)	-28.020* (-1.89)
Asset based loan	0.047** (2.60)	0.089*** (3.55)	45.130*** (3.42)	42.006** (2.65)
Second lien facility	0.006 (0.42)	-0.002 (-0.11)	-9.284 (-1.61)	-5.726 (-0.98)
Ln(Firm age)	-0.010* (-1.76)	-0.027*** (-3.61)	-4.126 (-1.27)	-1.781 (-0.36)
Industry effects	Yes	Yes	Yes	Yes
Year-quarter x rating effects	Yes	Yes	Yes	Yes
Observations	2186	2279	2186	2279
Weak identification test				
Kleibergen-Paap rk Wald F			6.596	12.960
Stock-Yogo critical values:				
10% maximal IV size:			16.38	16.38
15% maximal IV size:			8.96	8.96
20% maximal IV size:			6.66	6.66
Weak-IV robust inference:				
Anderson-Rubin χ^2			4.77**	4.02**
(p-value)			(0.0289)	(0.0450)
Stock-Wright LM S χ^2			5.01**	4.70**
(p-value)			(0.0252)	(0.0302)

